

Data Analysis Report: Breast Cancer Data

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1 Introduction and Data Preparation

This report analyzes the Breast Cancer Wisconsin dataset.

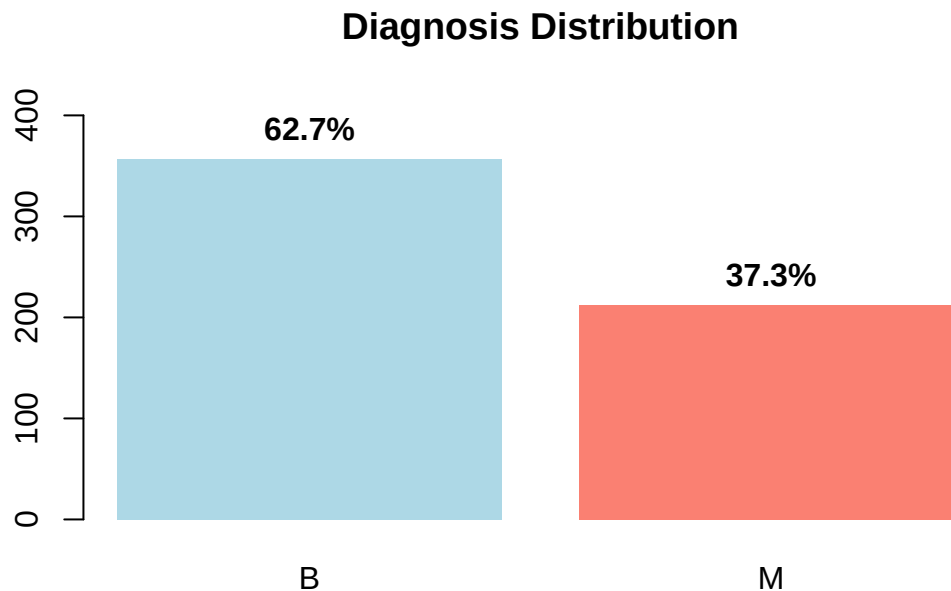
Ten real-valued features are computed for each cell nucleus:

- * radius (mean of distances from center to points on the perimeter)
- * texture (standard deviation of gray-scale values)
- * perimeter
- * area
- * smoothness (local variation in radius lengths)
- * compactness ($\text{perimeter}^2 / \text{area} - 1.0$)
- * concavity (severity of concave portions of the contour)
- * concave points (number of concave portions of the contour)
- * symmetry
- * fractal dimension ("coastline approximation")

The goal is to perform a complete statistical analysis including Univariate Analysis (distribution fitting), Principal Component Analysis (PCA), and Cluster Analysis.

1.1 Qualitative Variable Analysis

We start by analyzing the distribution of the target variable: Diagnosis (M = Malignant, B = Benign).



The dataset is slightly unbalanced, with a prevalence of Benign cases (62.7%).

1.2 Univariate Analysis and Goodness of Fit

In this section, we analyze each numerical variable individually. For each variable, we perform:

```

* Descriptive Statistics (Skewness, Kurtosis).

* Distribution Fitting (Exponential, Gamma, Inverse Gaussian, Lognormal, Weibull).

* Kolmogorov-Smirnov Test on the best fitting model (selected by AIC).

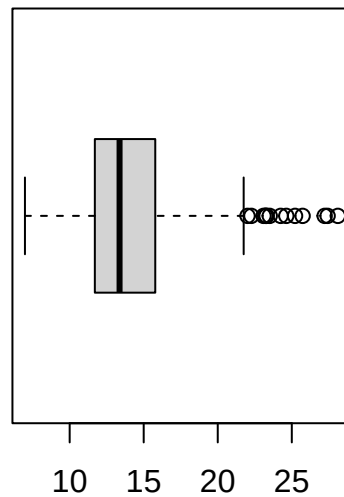
```

```

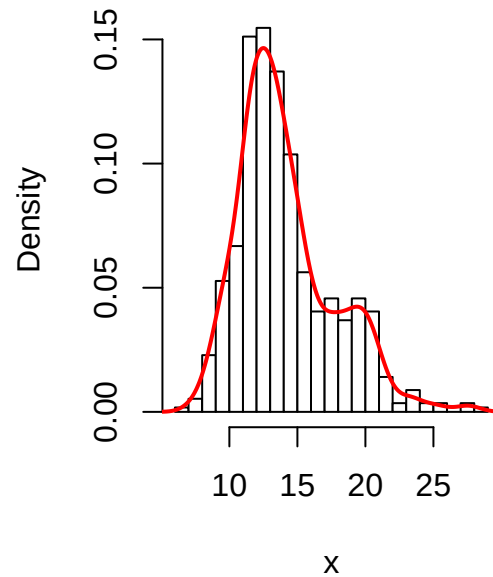
##
## ## Variable: radius_mean
##
## **Descriptive Statistics**
##
##
## |      Mean|      SD| Skewness| Kurtosis|
## |-----:|-----:|-----:|-----:|
## | 14.1273| 3.524|  0.9374|  0.8141|

```

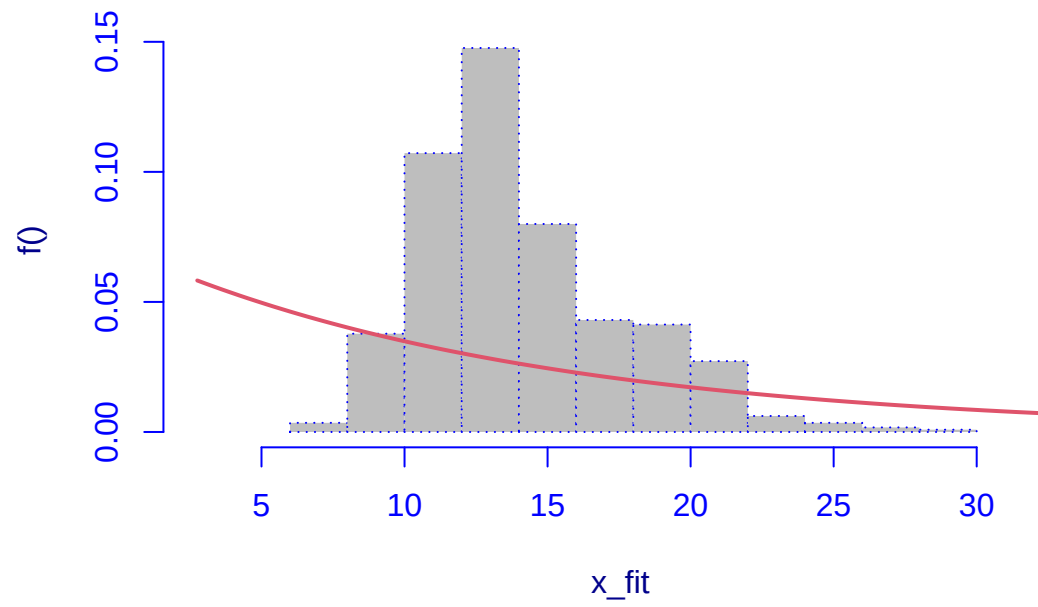
Boxplot



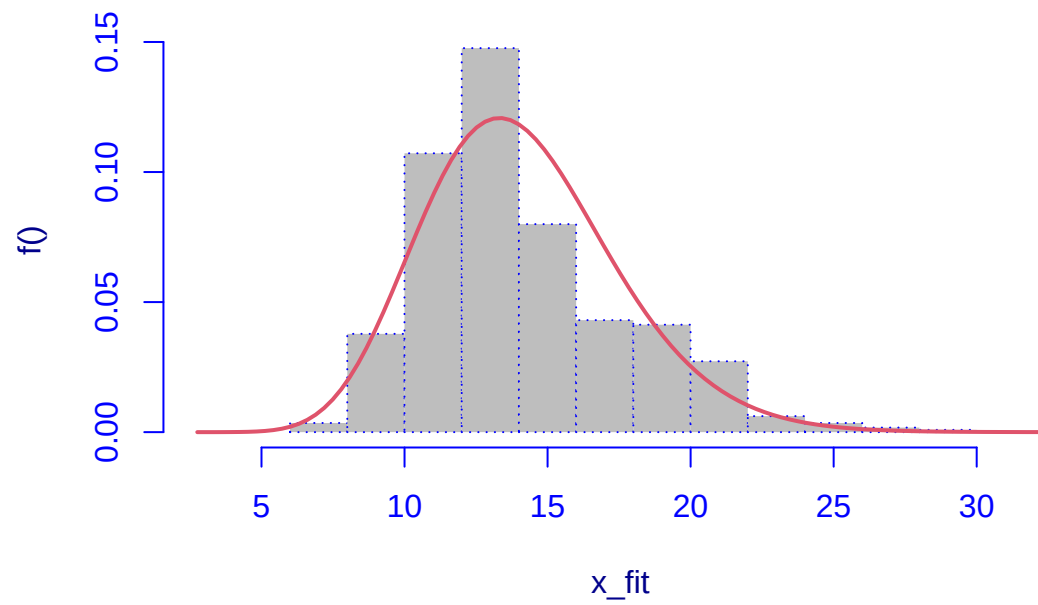
Histogram & Density



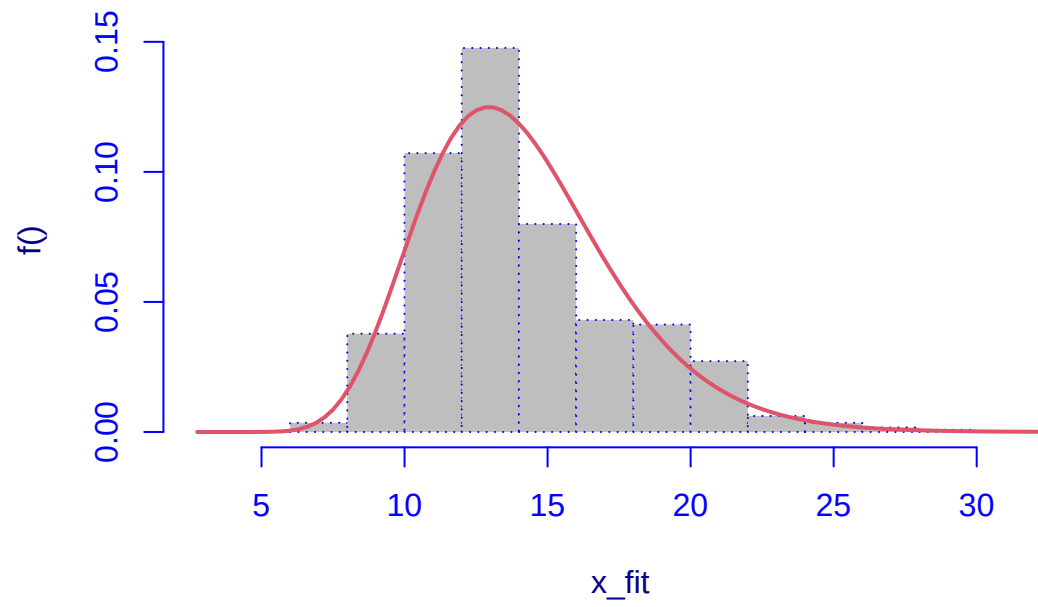
The x_{fit} and the fitted EXP distribution



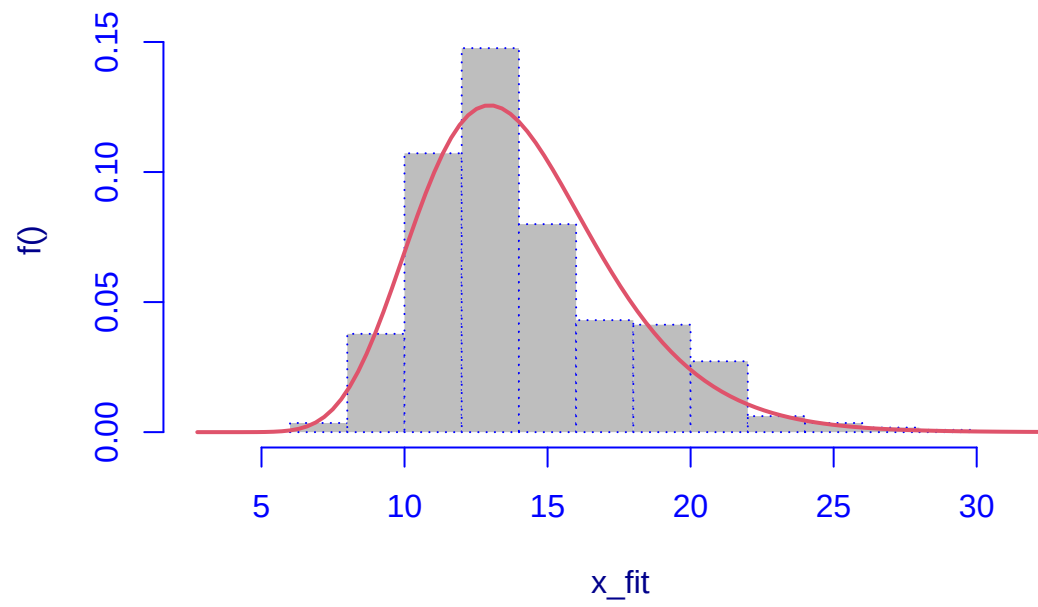
The x_{fit} and the fitted GA distribution



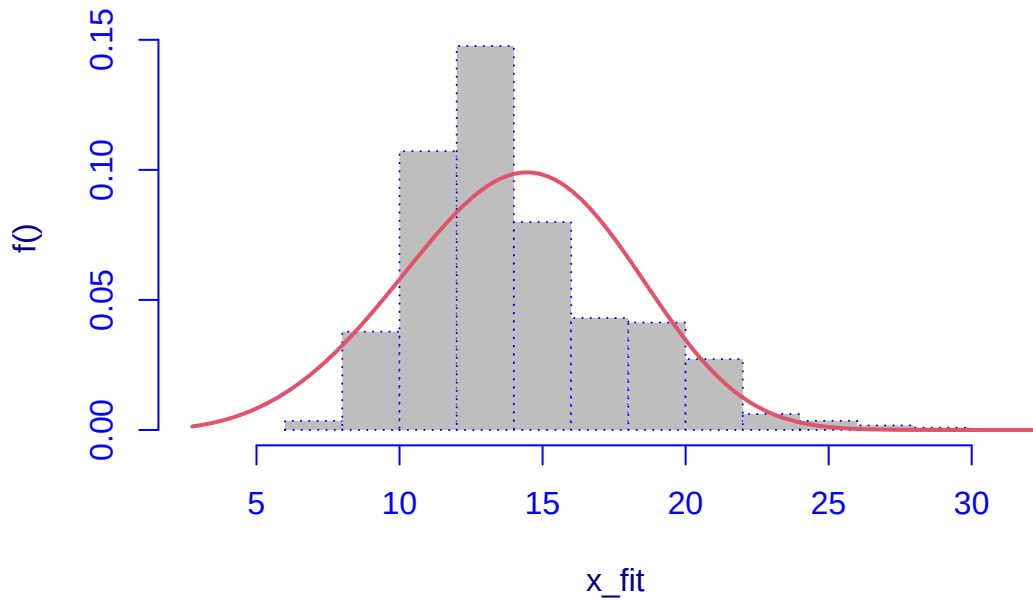
The x_{fit} and the fitted IG distribution



The x_{fit} and the fitted LOGNO distribution



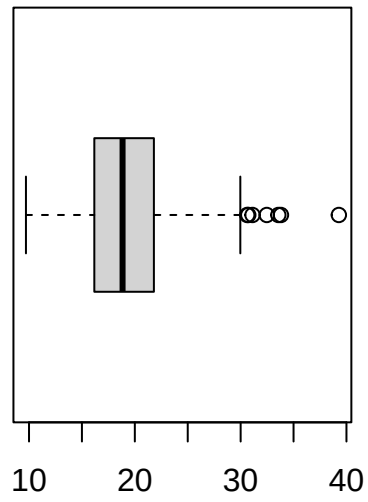
The x_{fit} and the fitted WEI distribution



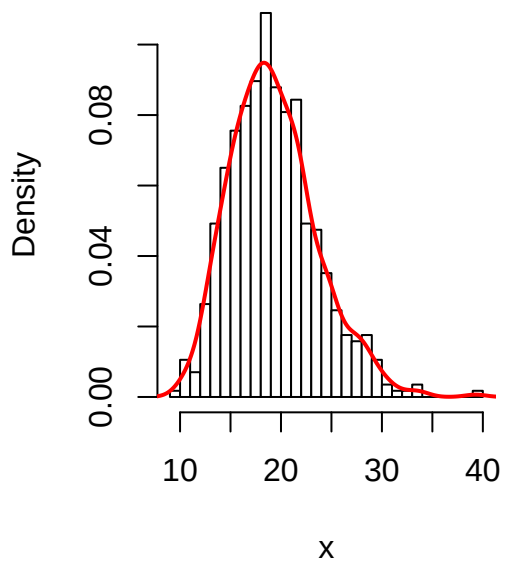
```
##
## **Model Comparison (Sorted by AIC)**
##
##
## |Distribution |      AIC|      BIC|
## |:-----:|-----:|-----:|
## |IG          | 2964.99| 2973.68|
## |LOGNO       | 2965.64| 2974.33|
## |GA          | 2984.23| 2992.92|
## |WEI         | 3096.20| 3104.89|
## |EXP         | 4153.55| 4157.89|

##
## **Goodness of Fit (Kolmogorov-Smirnov)**
##
## - **Best Model**: IG
## - **p-value**: 0.01318
## - **Result**: Reject H0. Significant discrepancy detected.
##
## \newpage
##
## ## Variable: texture_mean
##
## **Descriptive Statistics**
##
##
## |   Mean|   SD| Skewness| Kurtosis|
## |:-----:|-----:|-----:|-----:|
## | 19.2896| 4.301|   0.647|   0.728|
```

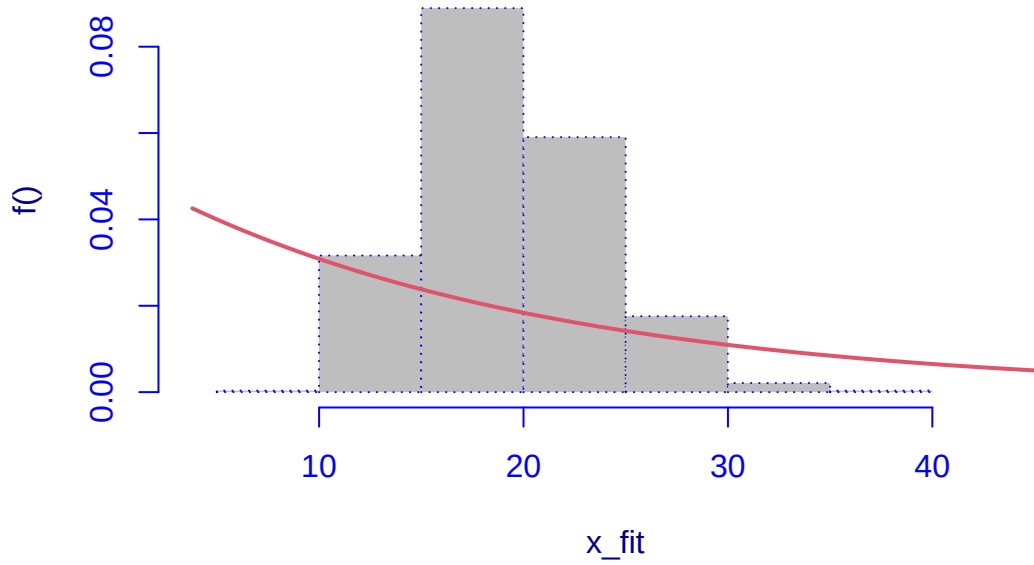
Boxplot



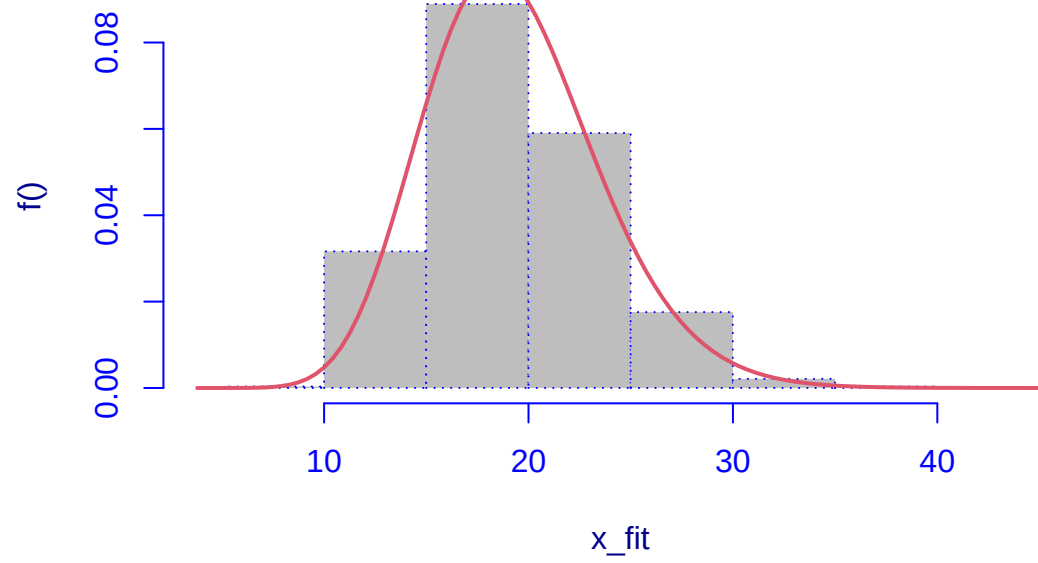
Histogram & Density



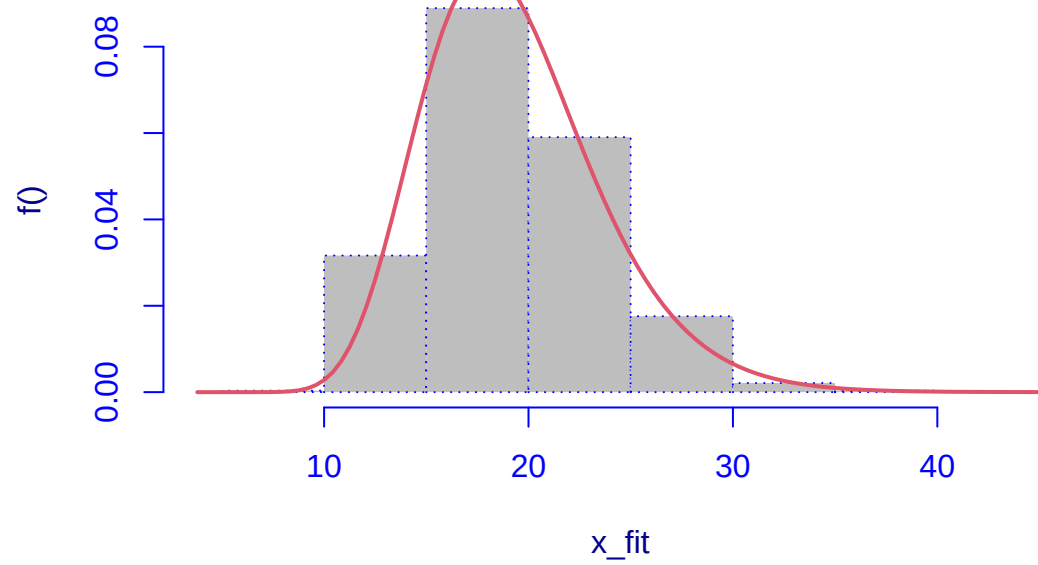
The x_{fit} and the fitted EXP distribution



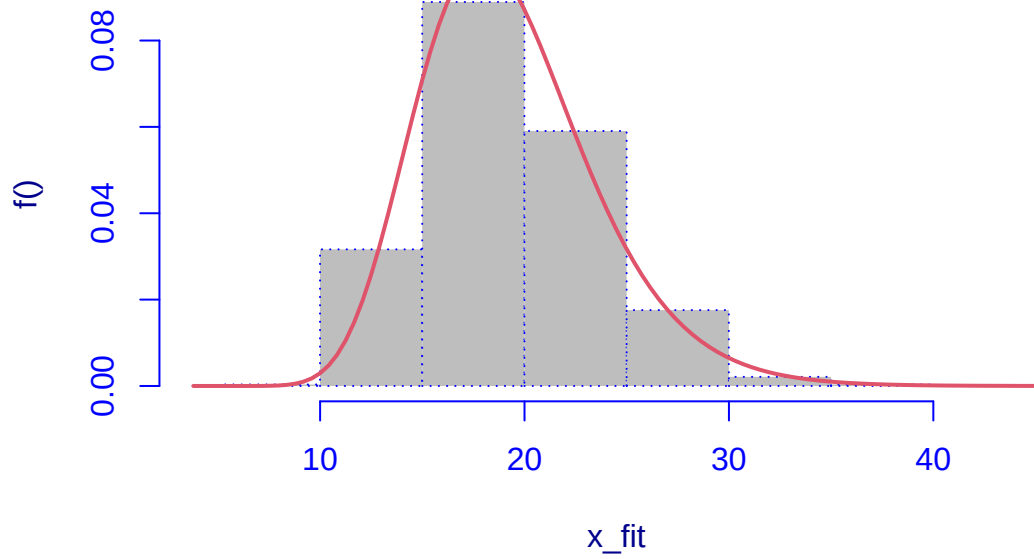
The x_{fit} and the fitted GA distribution



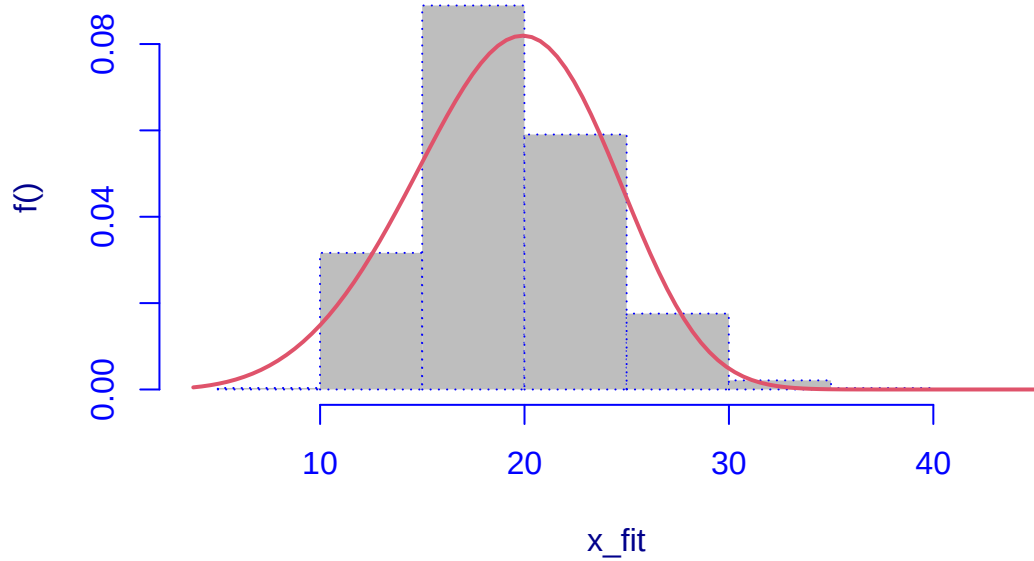
The x_{fit} and the fitted IG distribution



The x_{fit} and the fitted LOGNO distribution



The x_{fit} and the fitted WEI distribution

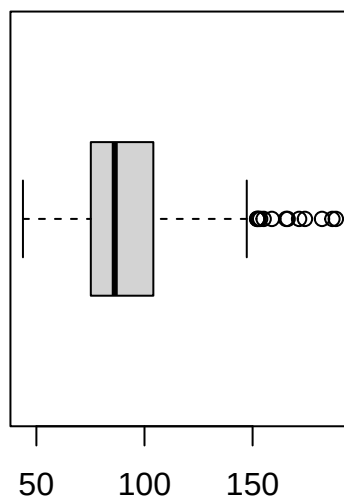


```
##
## **Model Comparison (Sorted by AIC)**
##
##
## |Distribution |      AIC|      BIC|
## |:-----:|-----:|-----:|
```

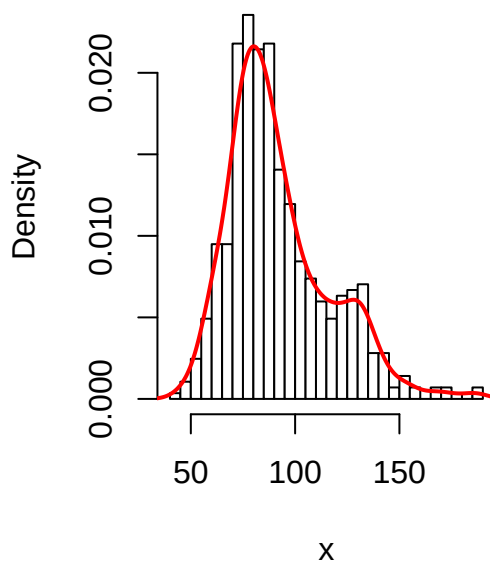
```
## |IG          | 3238.98| 3247.67|
## |LOGNO       | 3239.08| 3247.77|
## |GA          | 3242.95| 3251.63|
## |WEI         | 3326.80| 3335.49|
## |EXP         | 4507.99| 4512.33|

##
## **Goodness of Fit (Kolmogorov-Smirnov)**
##
## - **Best Model**: IG
## - **p-value**: 0.9938
## - **Result**: Do NOT reject H0. The model is a good fit.
##
## \newpage
##
## ## Variable: perimeter_mean
##
## **Descriptive Statistics**
##
##
## | Mean| SD| Skewness| Kurtosis|
## |-----:|-----:|-----:|-----:|
## | 91.969| 24.299| 0.9854| 0.9393|
```

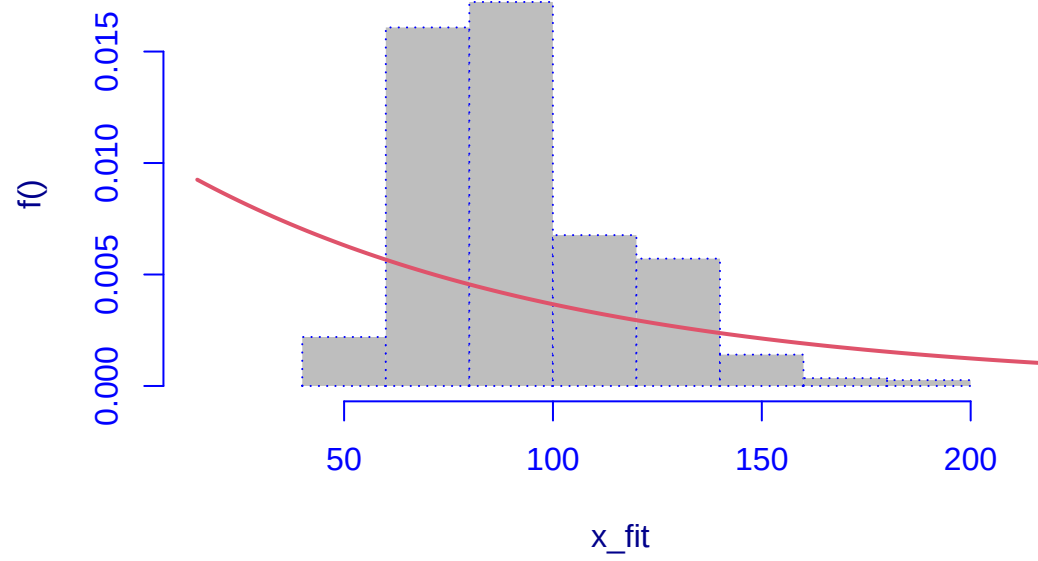
Boxplot



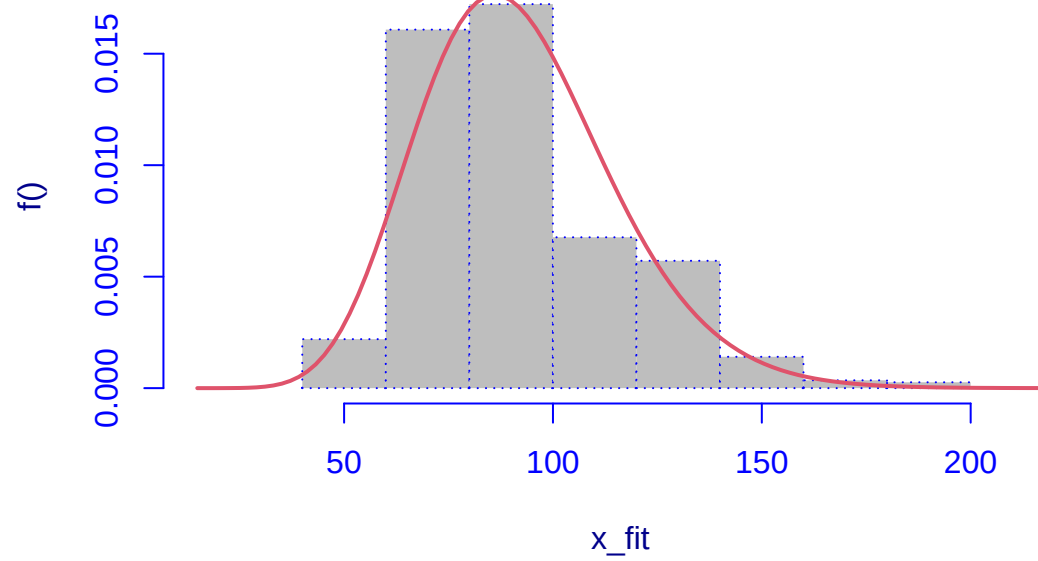
Histogram & Density



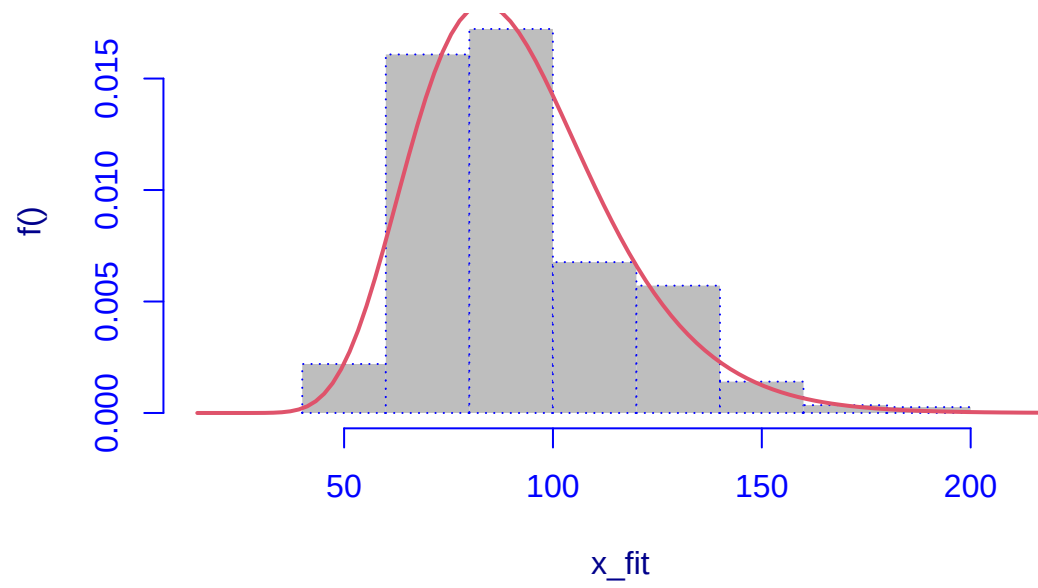
The x_{fit} and the fitted EXP distribution



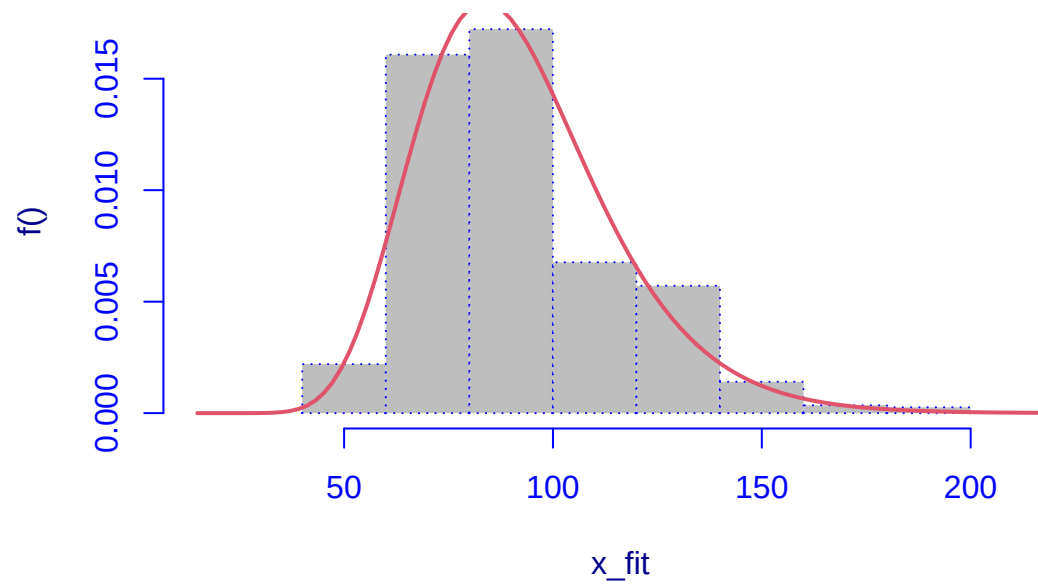
The x_{fit} and the fitted GA distribution



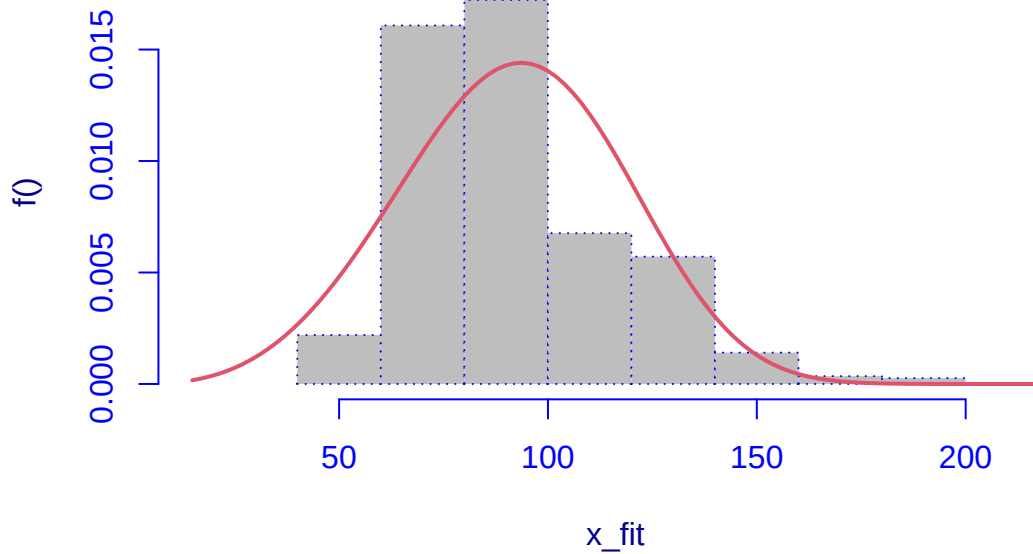
The x_{fit} and the fitted IG distribution



The x_{fit} and the fitted LOGNO distribution



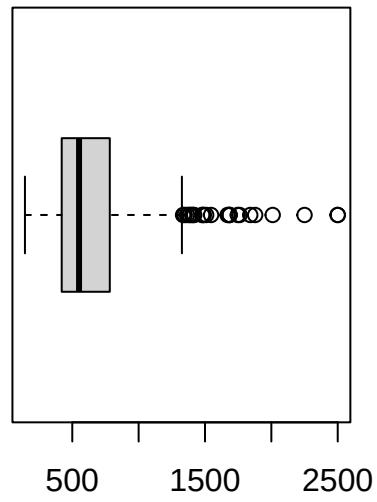
The x_{fit} and the fitted WEI distribution



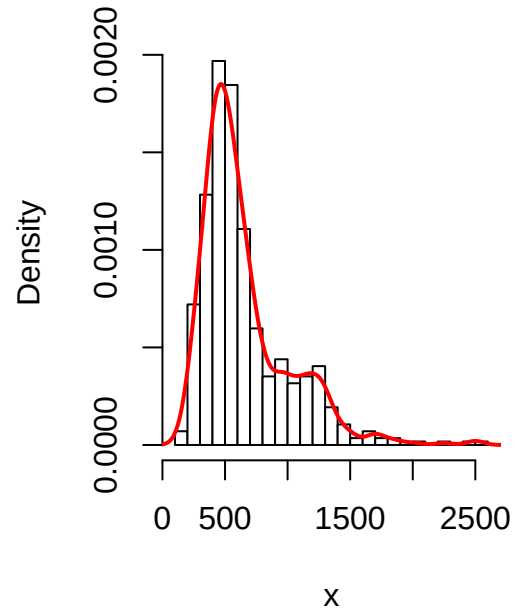
```
##
## **Model Comparison (Sorted by AIC)**
##
##
## |Distribution |      AIC|      BIC|
## |:-----:|:-----:|:-----:|
## |IG          | 5152.99| 5161.67|
## |LOGNO       | 5153.75| 5162.44|
## |GA          | 5174.33| 5183.02|
## |WEI         | 5286.53| 5295.22|
## |EXP         | 6285.41| 6289.76|

##
## **Goodness of Fit (Kolmogorov-Smirnov)**
##
## - **Best Model**: IG
## - **p-value**: 0.003872
## - **Result**: Reject H0. Significant discrepancy detected.
##
## \newpage
##
## ## Variable: area_mean
##
## **Descriptive Statistics**
##
##
## |      Mean|      SD| Skewness| Kurtosis|
## |:-----:|:-----:|:-----:|:-----:|
## | 654.8891| 351.9141|  1.6371|  3.5865|
```

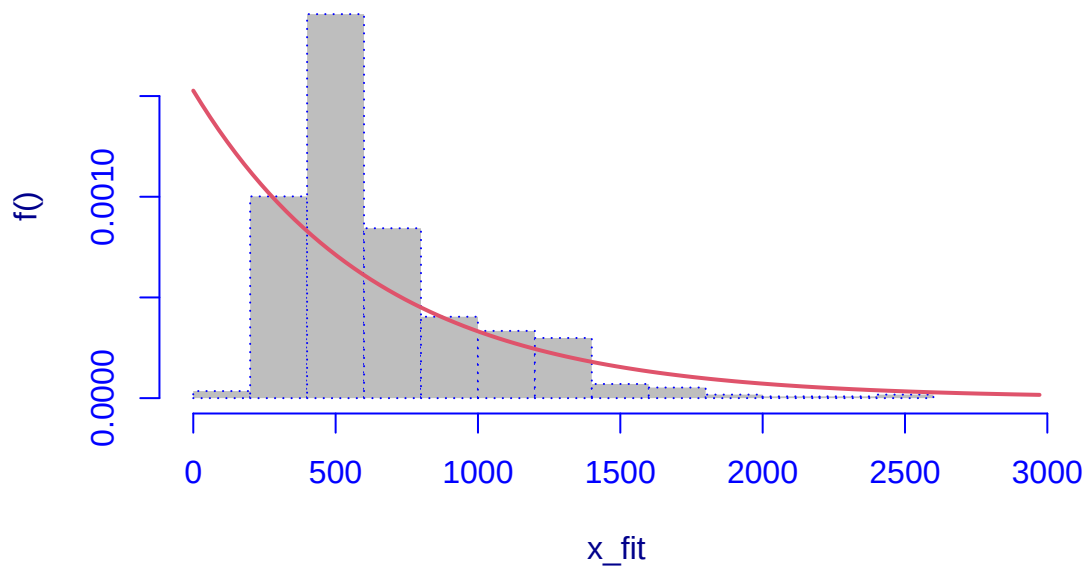
Boxplot



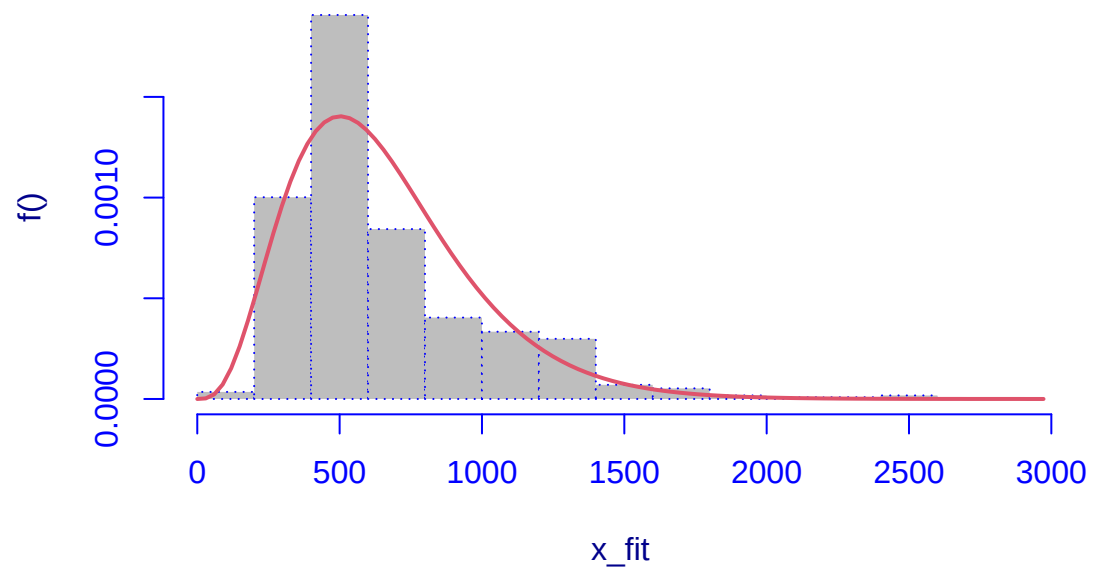
Histogram & Density



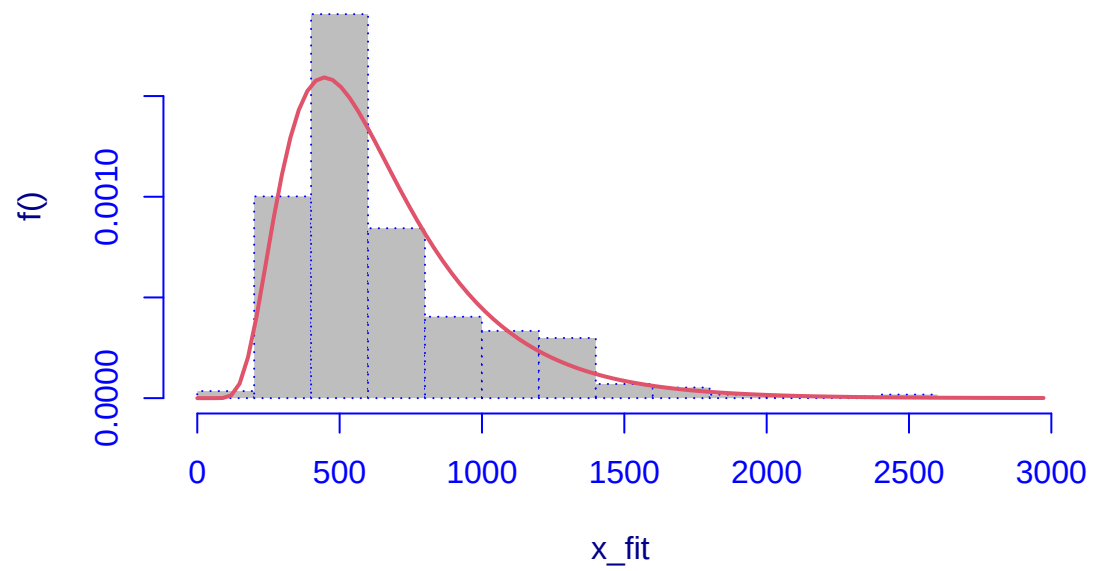
The x_{fit} and the fitted EXP distribution



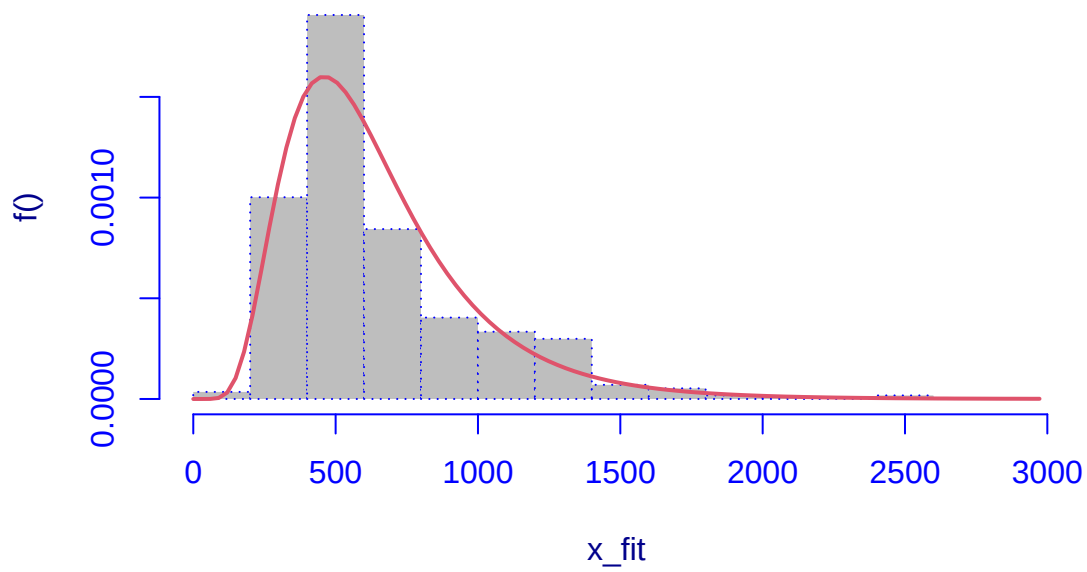
The x_{fit} and the fitted GA distribution



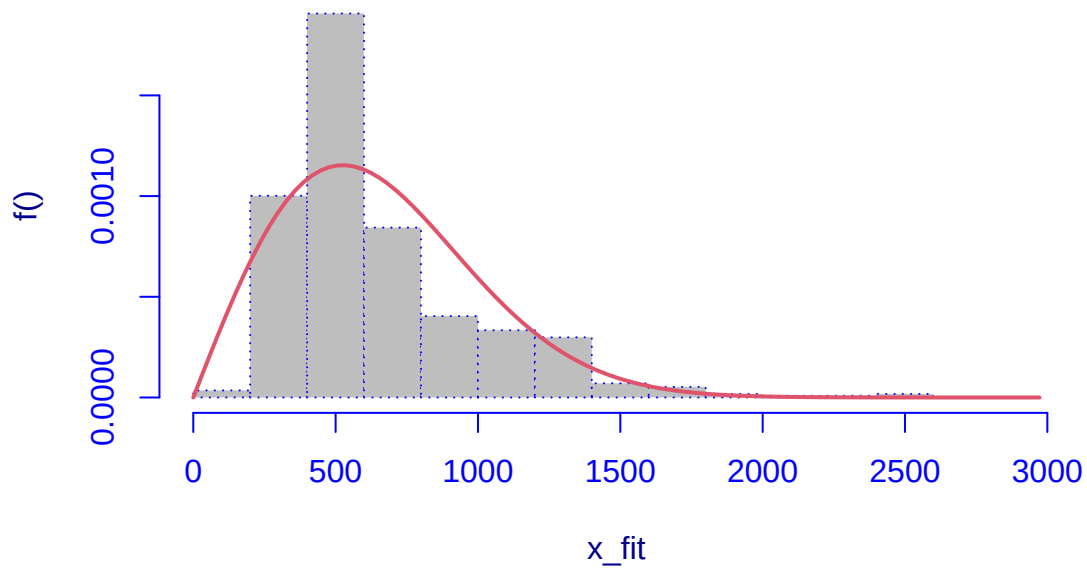
The x_{fit} and the fitted IG distribution



The x_{fit} and the fitted LOGNO distribution



The x_{fit} and the fitted WEI distribution



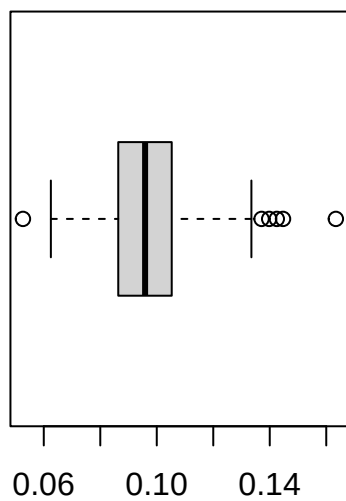
```
##
## **Model Comparison (Sorted by AIC)**
##
##
## |Distribution |      AIC|      BIC|
## |:-----:|-----:|-----:|
```



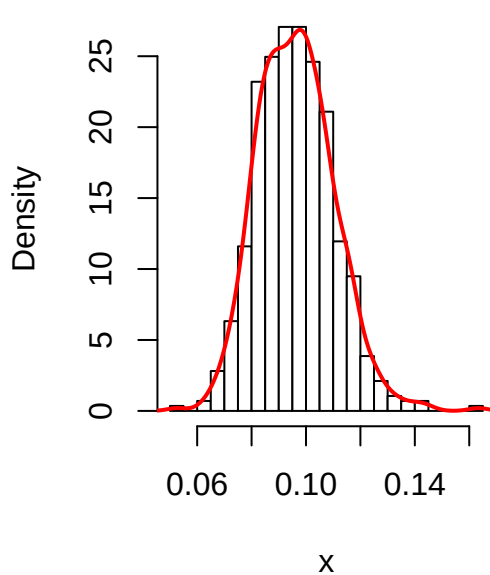
```
## |IG          | 8028.18| 8036.87|
## |LOGNO       | 8031.22| 8039.90|
## |GA          | 8076.59| 8085.28|
## |WEI         | 8158.78| 8167.47|
## |EXP         | 8519.32| 8523.67|

##
## **Goodness of Fit (Kolmogorov-Smirnov)**
##
## - **Best Model**: IG
## - **p-value**: 0.01804
## - **Result**: Reject H0. Significant discrepancy detected.
##
## \newpage
##
## ## Variable: smoothness_mean
##
## **Descriptive Statistics**
##
##
## | Mean| SD| Skewness| Kurtosis|
## |-----:|-----:|-----:|-----:|
## | 0.0964| 0.0141| 0.4539| 0.8245|
```

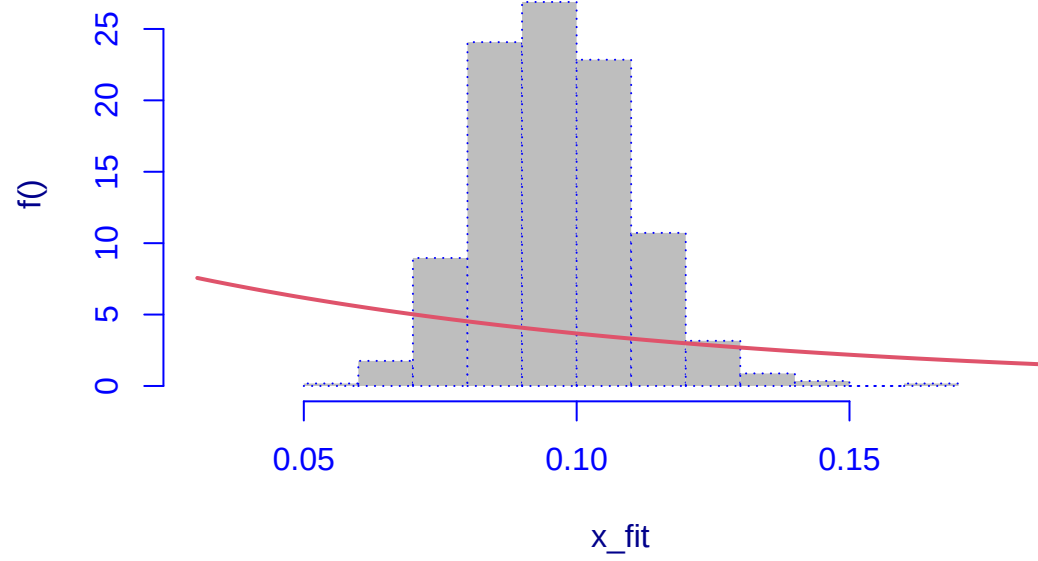
Boxplot



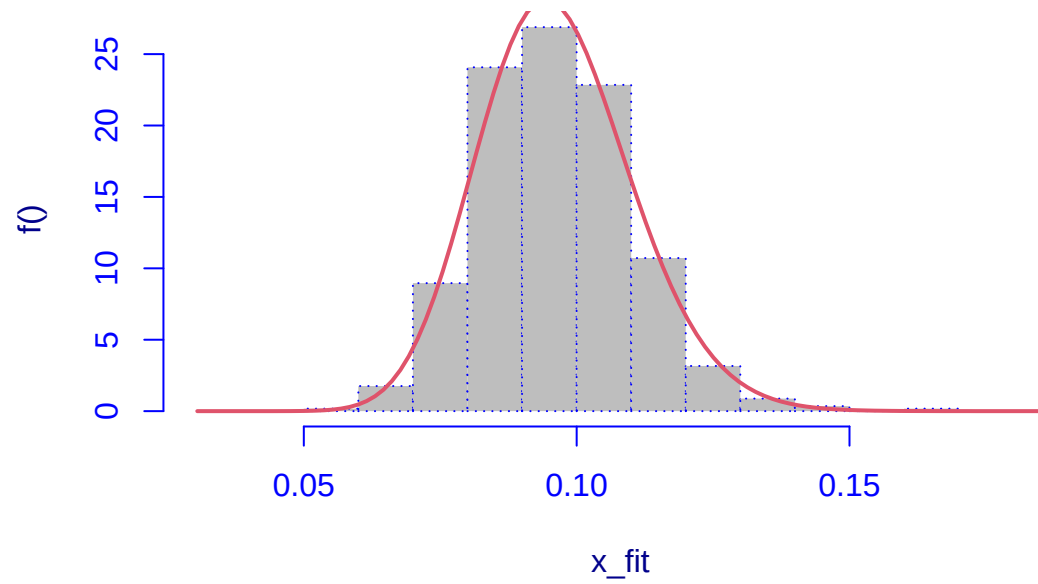
Histogram & Density



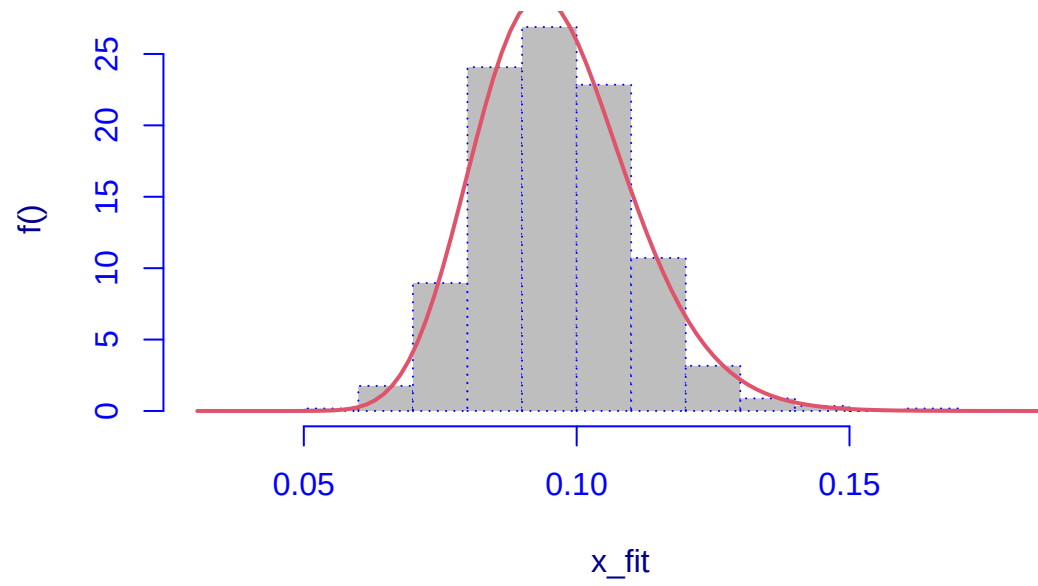
The x_{fit} and the fitted EXP distribution



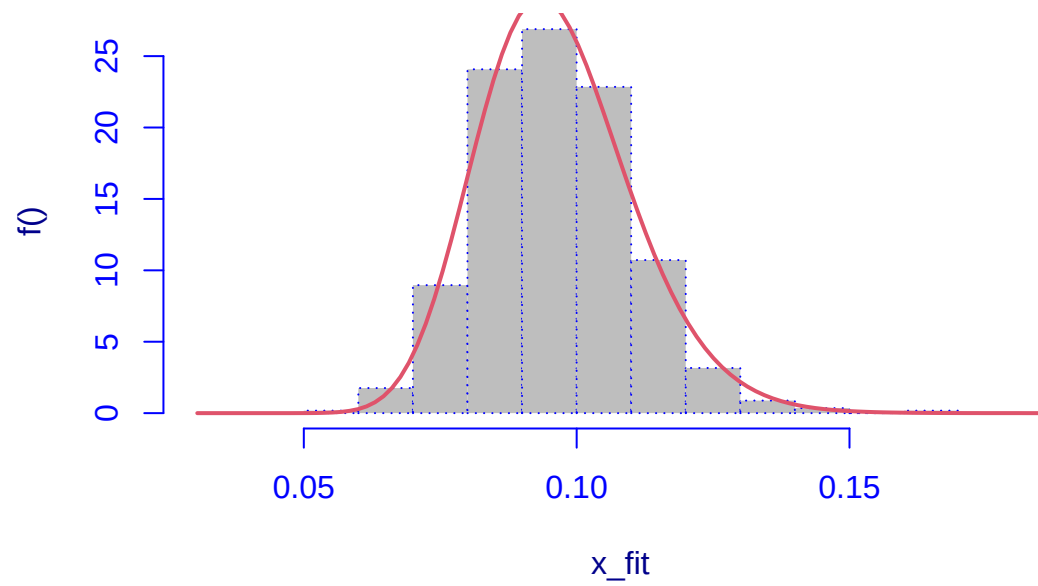
The x_{fit} and the fitted GA distribution



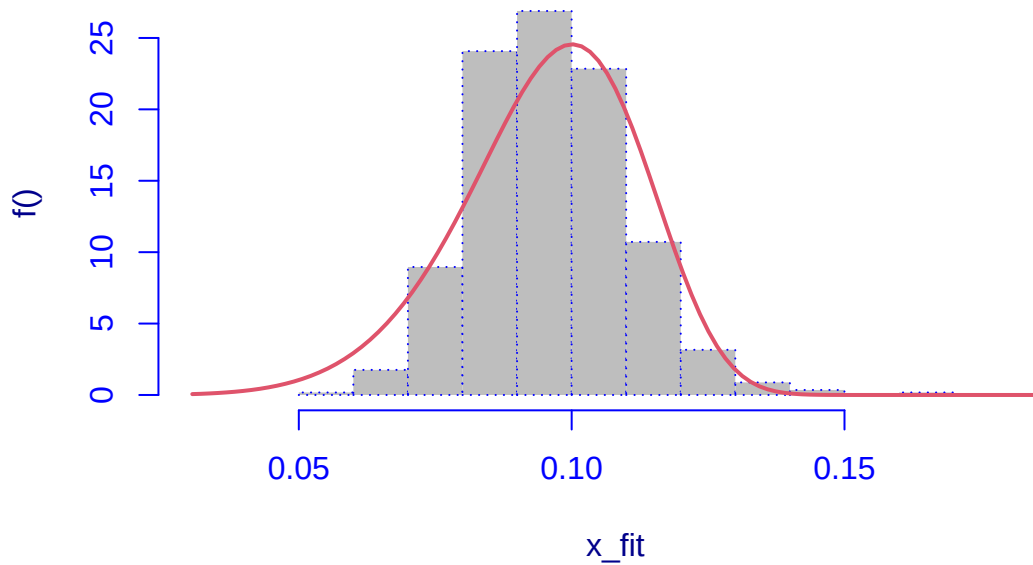
The x_{fit} and the fitted IG distribution



The x_{fit} and the fitted LOGNO distribution



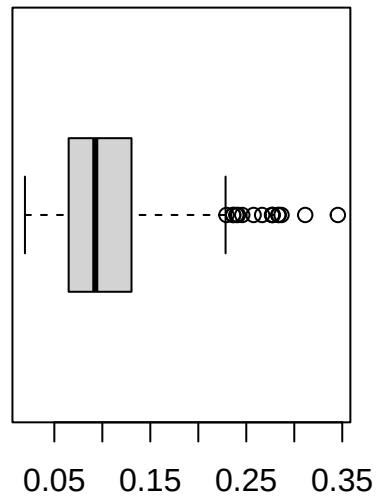
The x_{fit} and the fitted WEI distribution



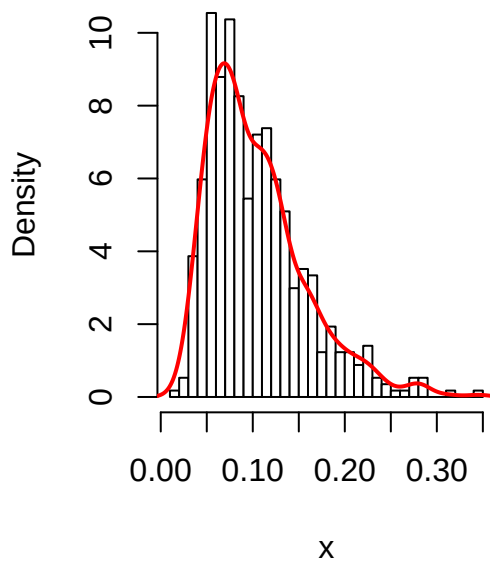
```
##
## **Model Comparison (Sorted by AIC)**
##
##
## |Distribution |      AIC|      BIC|
## |:-----:|:-----:|:-----:|
## |LOGNO      | -3249.81| -3241.12|
## |IG         | -3249.40| -3240.71|
## |GA         | -3249.31| -3240.62|
## |WEI        | -3151.79| -3143.11|
## |EXP        | -1522.53| -1518.19|

##
## **Goodness of Fit (Kolmogorov-Smirnov)**
##
## - **Best Model**: LOGNO
## - **p-value**: 0.8439
## - **Result**: Do NOT reject H0. The model is a good fit.
##
## \newpage
##
## ## Variable: compactness_mean
##
## **Descriptive Statistics**
##
##
## | Mean|      SD| Skewness| Kurtosis|
## |:-----:|:-----:|:-----:|:-----:|
## | 0.1043| 0.0528| 1.1839| 1.6089|
```

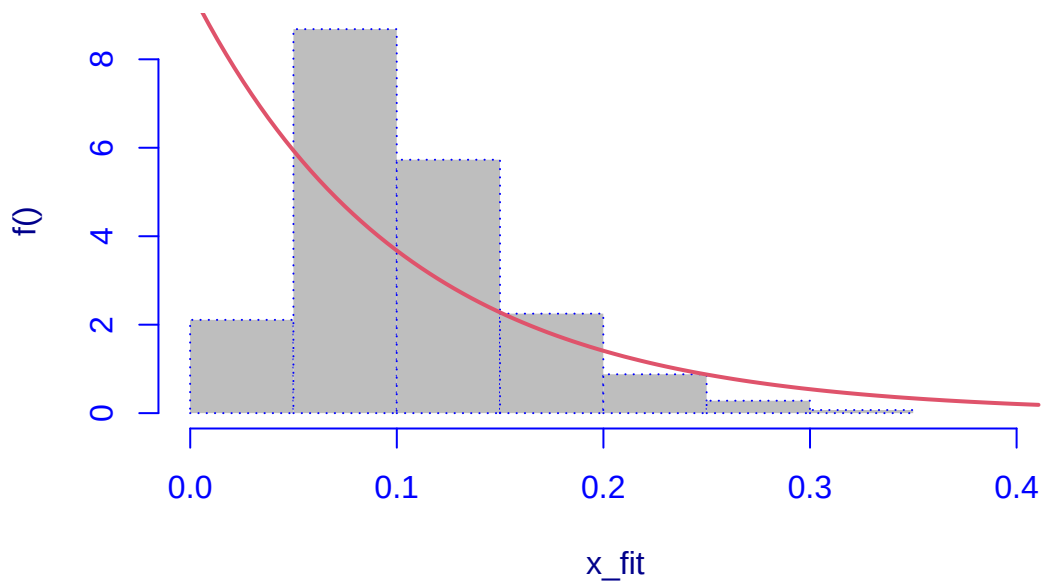
Boxplot



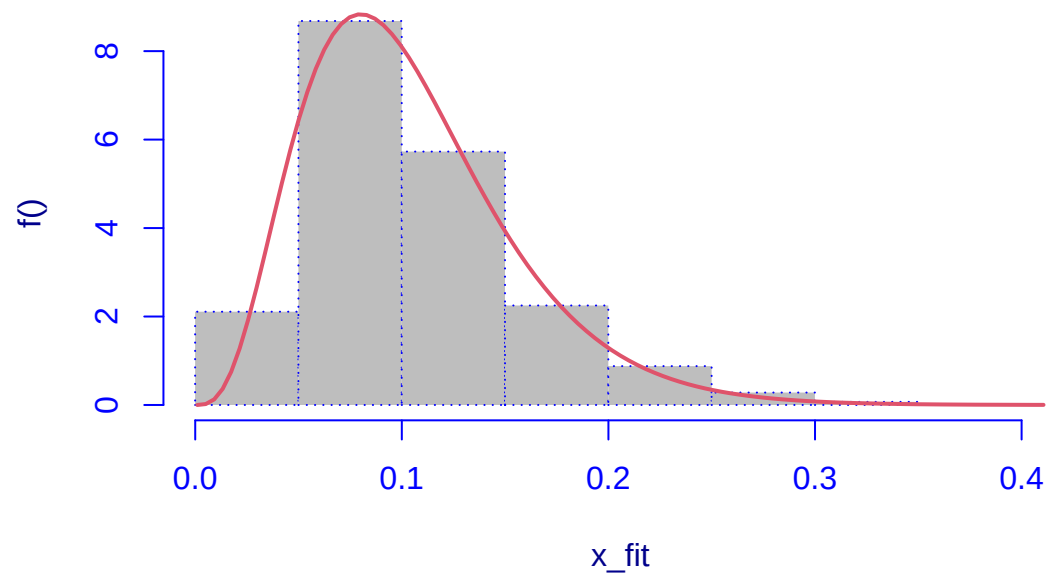
Histogram & Density



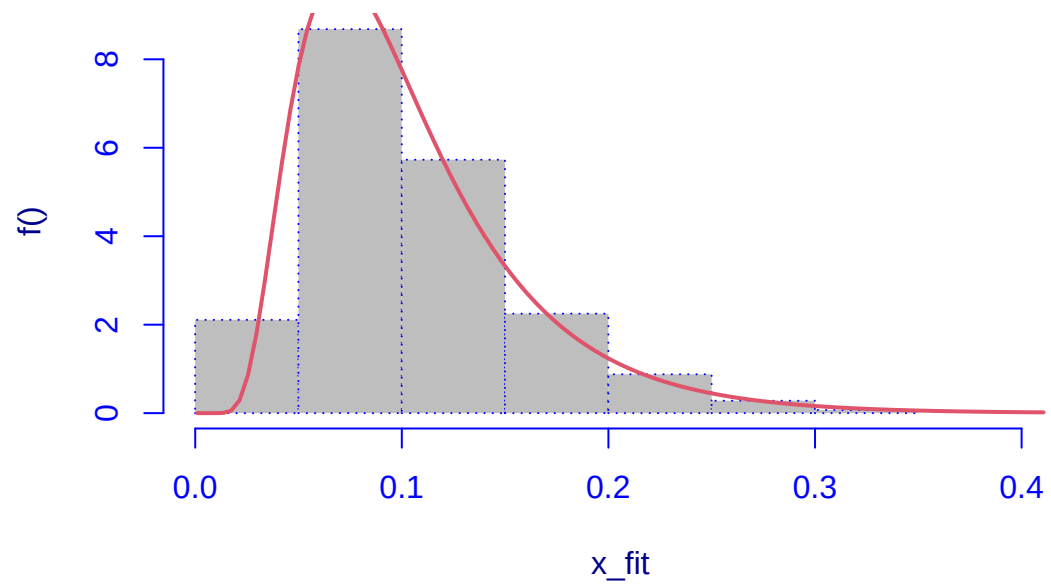
The x_{fit} and the fitted EXP distribution



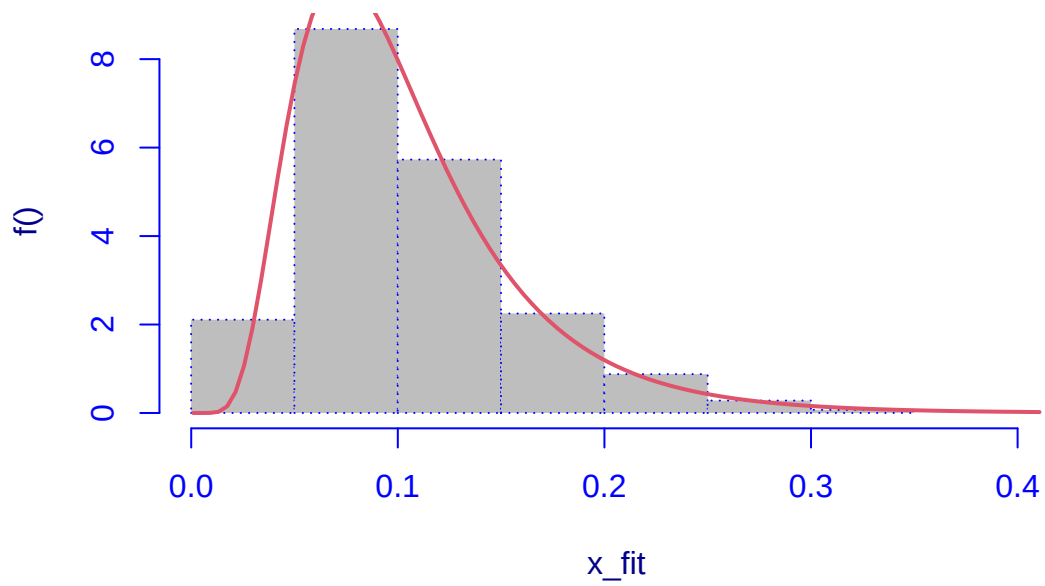
The x_{fit} and the fitted GA distribution



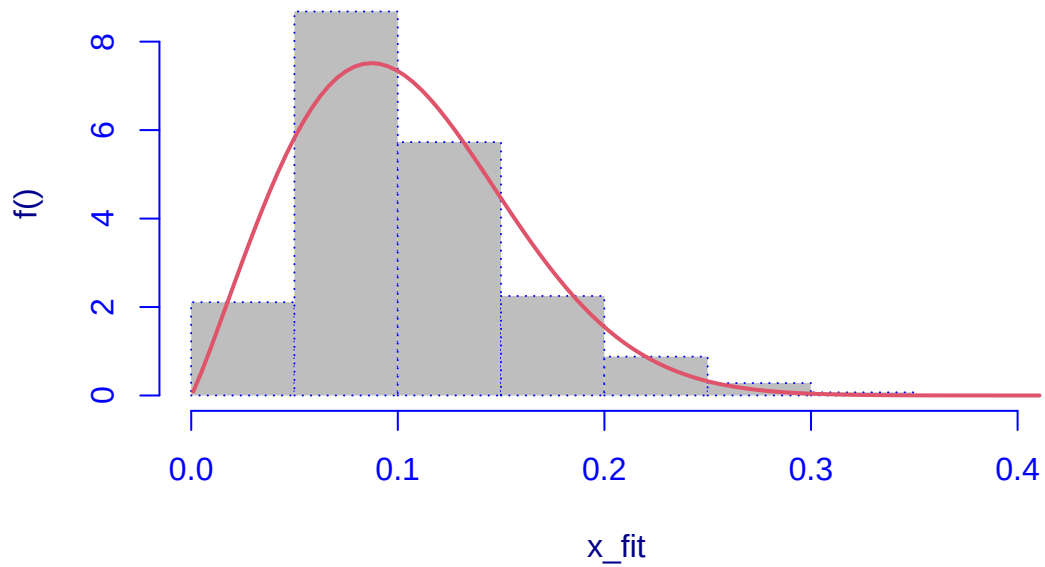
The x_{fit} and the fitted IG distribution



The x_{fit} and the fitted LOGNO distribution



The x_{fit} and the fitted WEI distribution

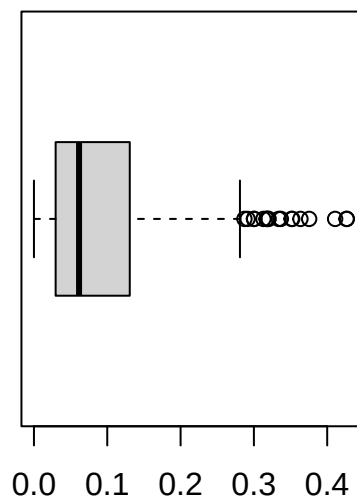


```
##
## **Model Comparison (Sorted by AIC)**
##
##
## |Distribution |      AIC|      BIC|
## |:-----:|-----:|-----:|
```

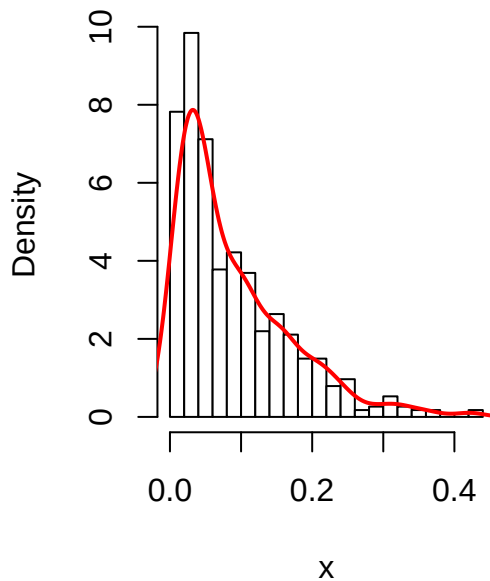
```
## |IG          | -1895.14| -1886.45|
## |LOGNO       | -1892.76| -1884.07|
## |GA          | -1877.92| -1869.23|
## |WEI         | -1825.50| -1816.81|
## |EXP         | -1431.98| -1427.64|

##
## **Goodness of Fit (Kolmogorov-Smirnov)**
##
## - **Best Model**: IG
## - **p-value**: 0.5696
## - **Result**: Do NOT reject H0. The model is a good fit.
##
## \newpage
##
## ## Variable: concavity_mean
##
## **Descriptive Statistics**
##
##
## | Mean| SD| Skewness| Kurtosis|
## |-----:|-----:|-----:|-----:|
## | 0.0888| 0.0797| 1.3938| 1.9531|
```

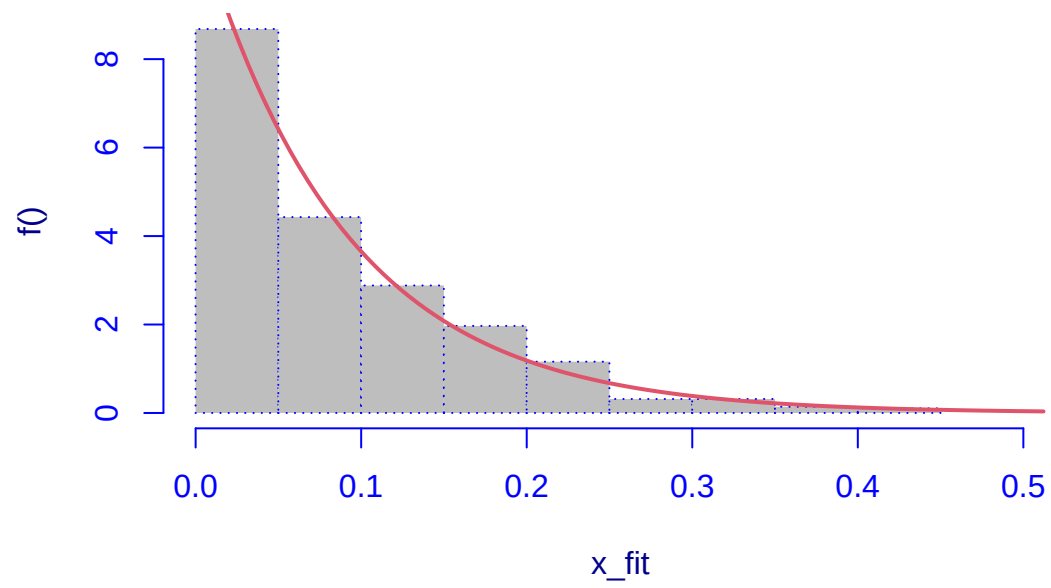
Boxplot



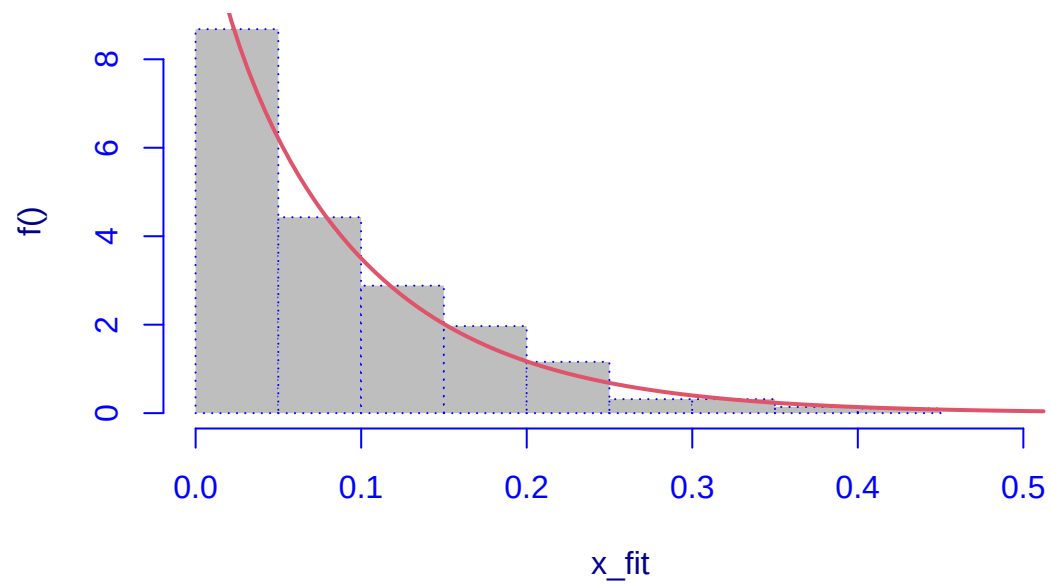
Histogram & Density



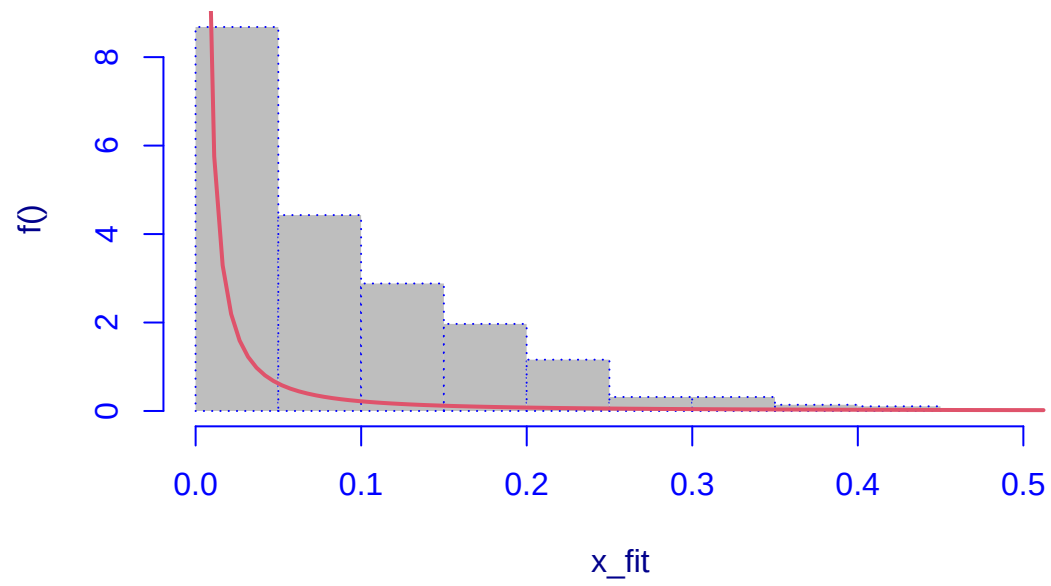
The x_{fit} and the fitted EXP distribution



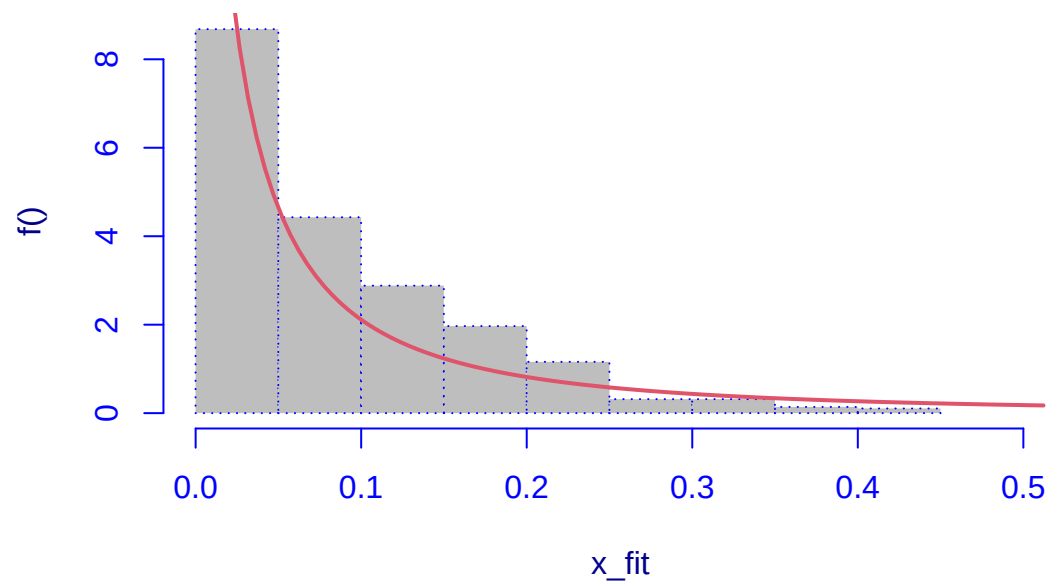
The x_{fit} and the fitted GA distribution



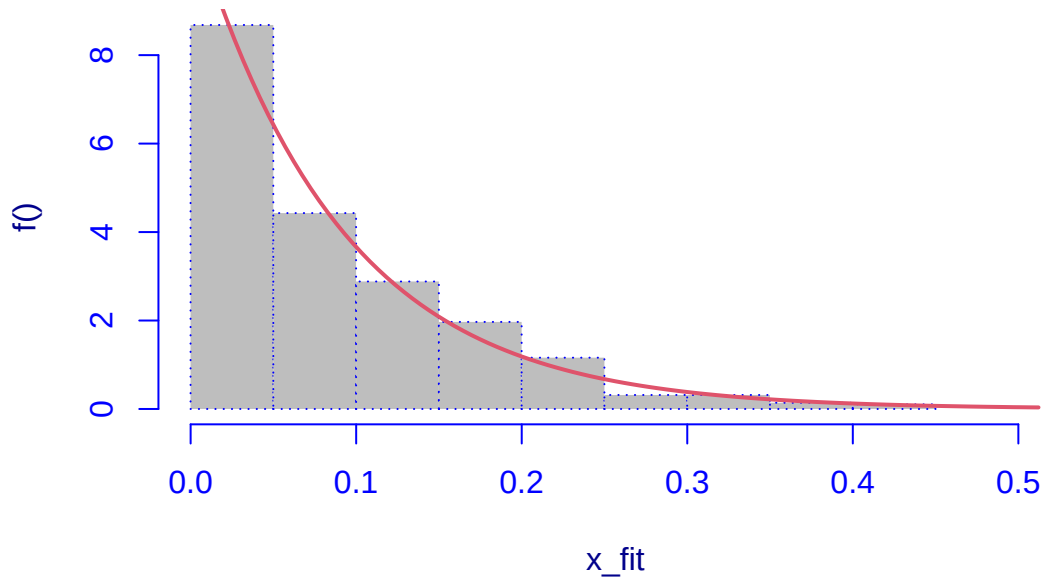
The x_{fit} and the fitted IG distribution



The x_{fit} and the fitted LOGNO distribution



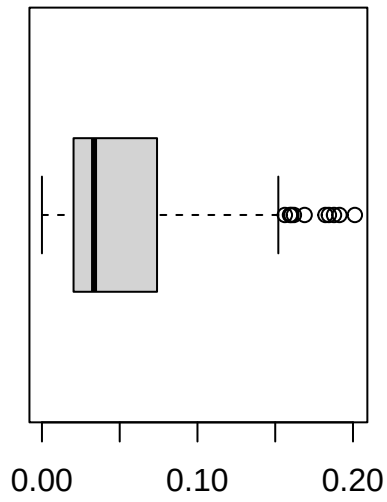
The x_{fit} and the fitted WEI distribution



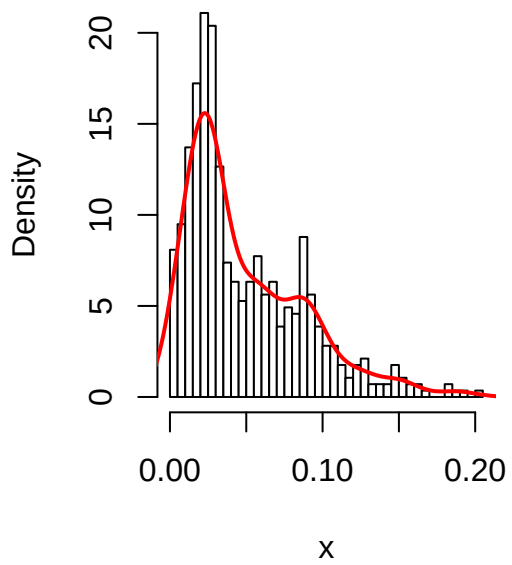
```
##
## **Model Comparison (Sorted by AIC)**
##
##
## |Distribution |      AIC|      BIC|
## |:-----:|:-----:|:-----:|
## |GA          | -1615.52| -1606.83|
## |EXP         | -1615.44| -1611.09|
## |WEI         | -1613.44| -1604.76|
## |LOGNO       | -1234.23| -1225.54|
## |IG          |  1029.90|  1038.59|

##
## **Goodness of Fit (Kolmogorov-Smirnov)**
##
## - **Best Model**: GA
## - **p-value**: 0.004541
## - **Result**: Reject H0. Significant discrepancy detected.
##
## \newpage
##
## ## Variable: concave_points_mean
##
## **Descriptive Statistics**
##
##
## |  Mean|      SD| Skewness| Kurtosis|
## |:-----:|:-----:|:-----:|:-----:|
## | 0.0489| 0.0388|    1.165|    1.0325|
```

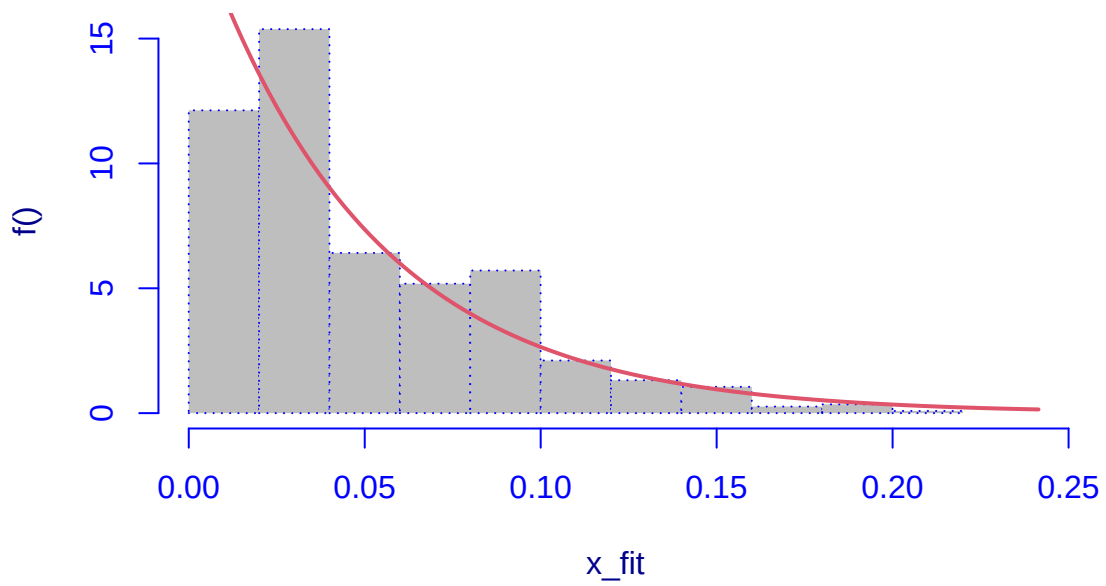
Boxplot



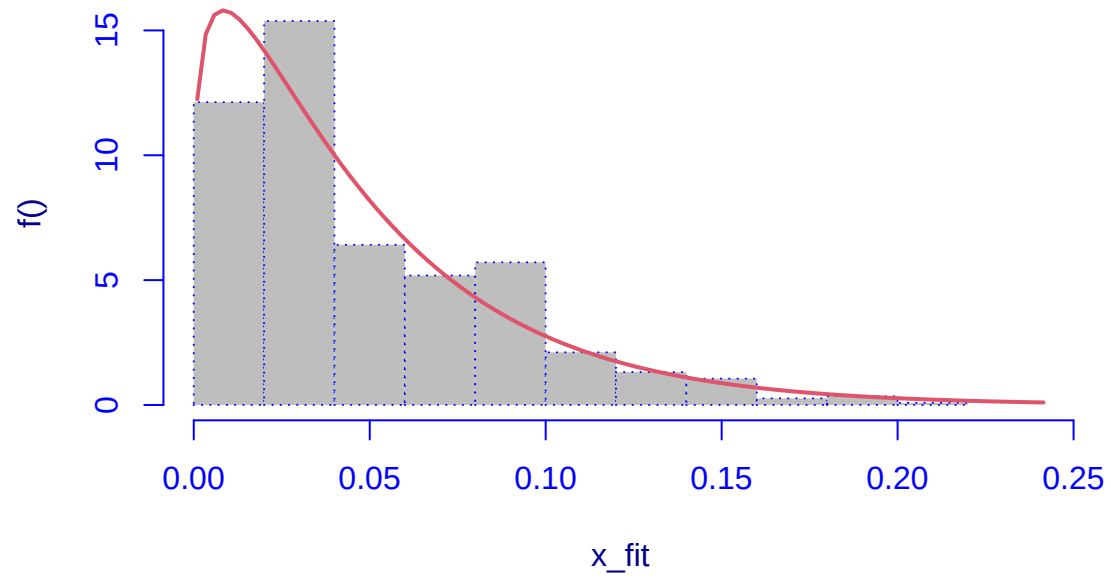
Histogram & Density



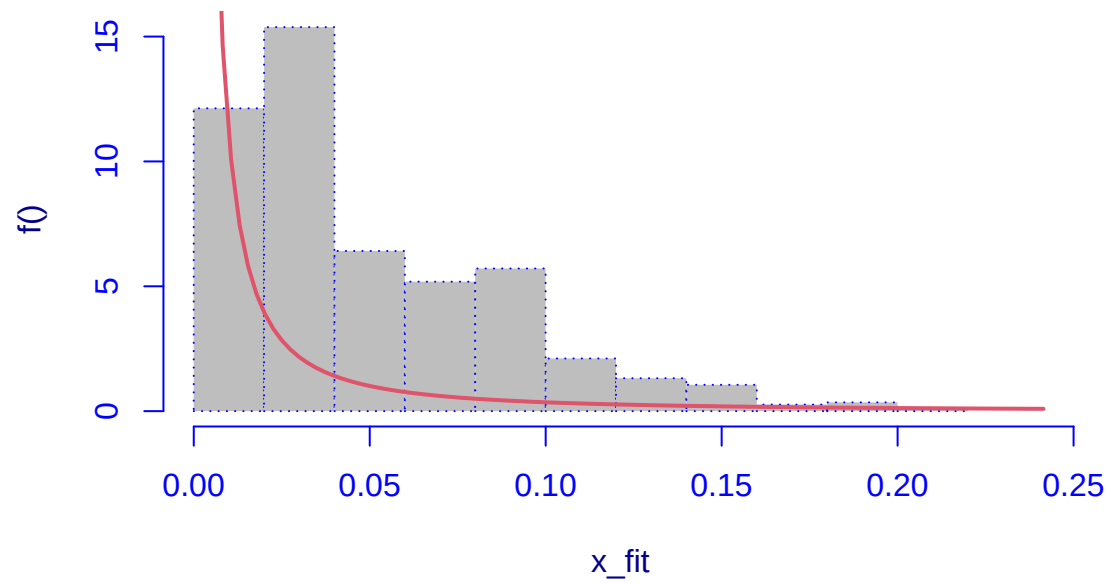
The x_{fit} and the fitted EXP distribution



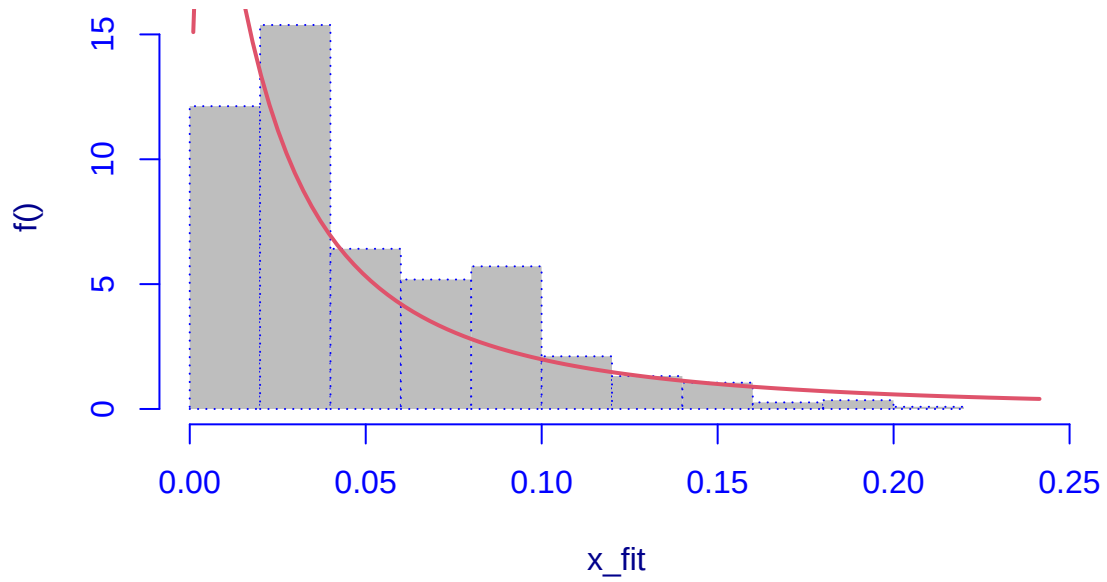
The x_{fit} and the fitted GA distribution



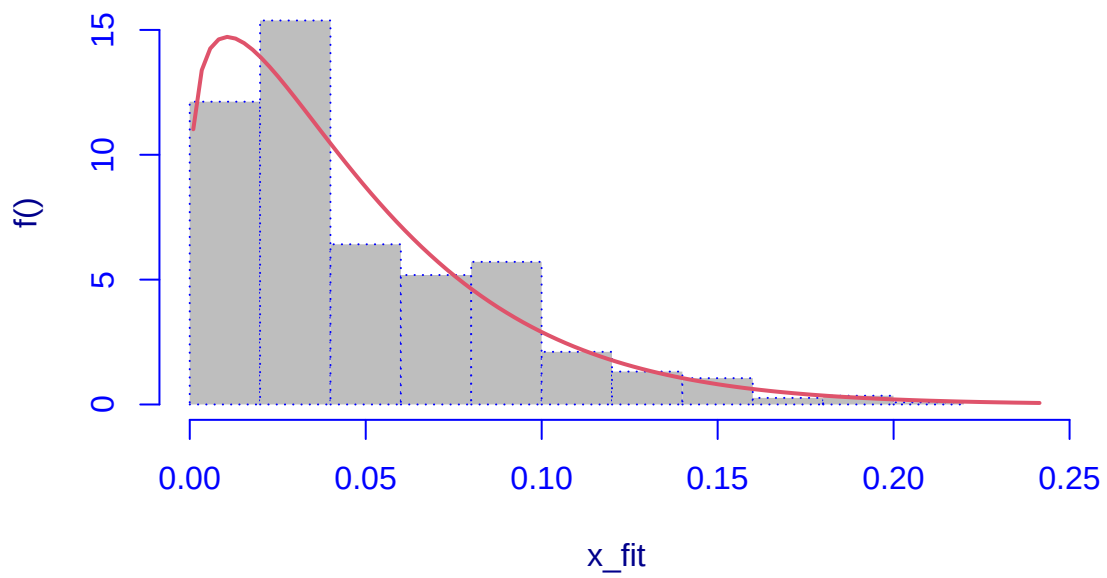
The x_{fit} and the fitted IG distribution



The x_{fit} and the fitted LOGNO distribution



The x_{fit} and the fitted WEI distribution

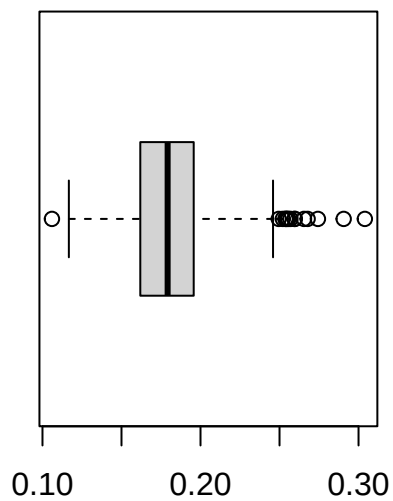


```
##
## **Model Comparison (Sorted by AIC)**
##
##
## |Distribution |      AIC|      BIC|
## |:-----:|-----:|-----:|
```

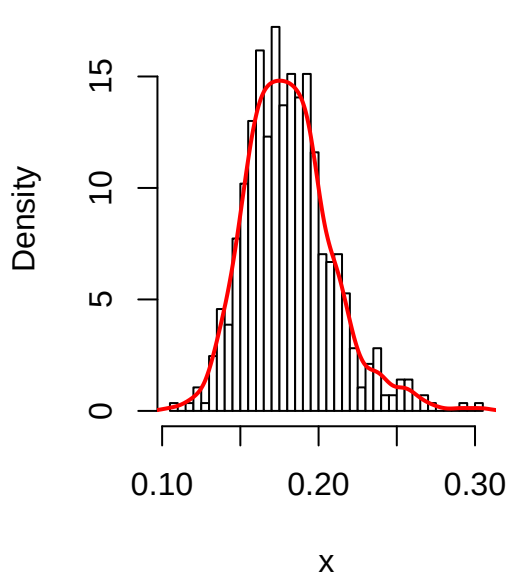
```
## |WEI          | -2315.12| -2306.43|
## |GA           | -2303.40| -2294.71|
## |EXP          | -2293.58| -2289.24|
## |LOGNO        | -1954.37| -1945.68|
## |IG           | -272.89| -264.21|

##
## **Goodness of Fit (Kolmogorov-Smirnov)**
##
## - **Best Model**: WEI
## - **p-value**: 0.04021
## - **Result**: Reject H0. Significant discrepancy detected.
##
## \newpage
##
## ## Variable: symmetry_mean
##
## **Descriptive Statistics**
##
##
## | Mean| SD| Skewness| Kurtosis|
## |-----:|-----:|-----:|-----:|
## | 0.1812| 0.0274| 0.7218| 1.2511|
```

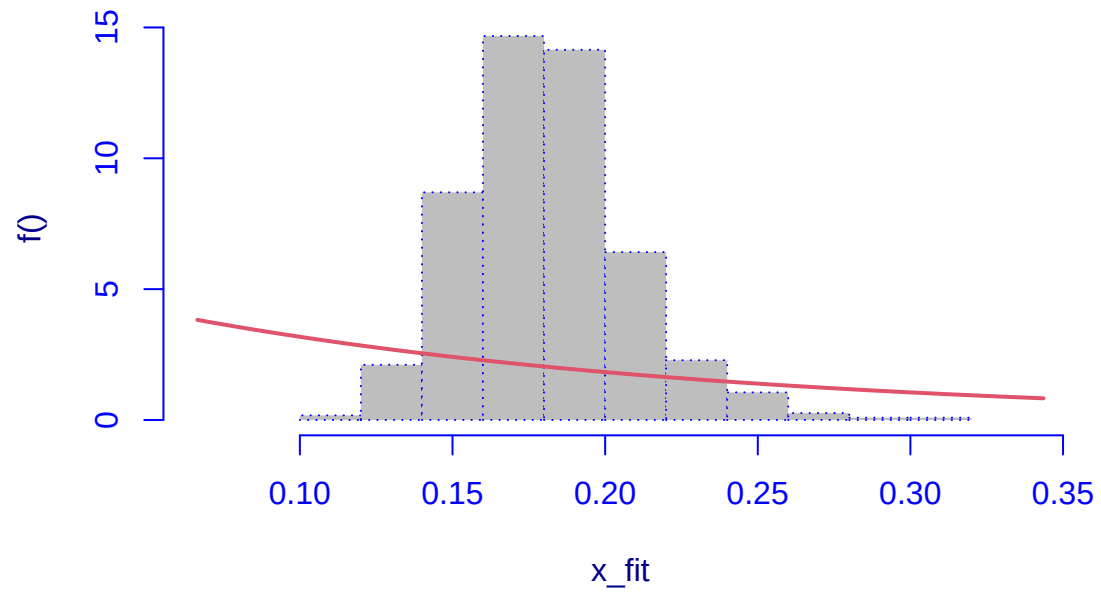
Boxplot



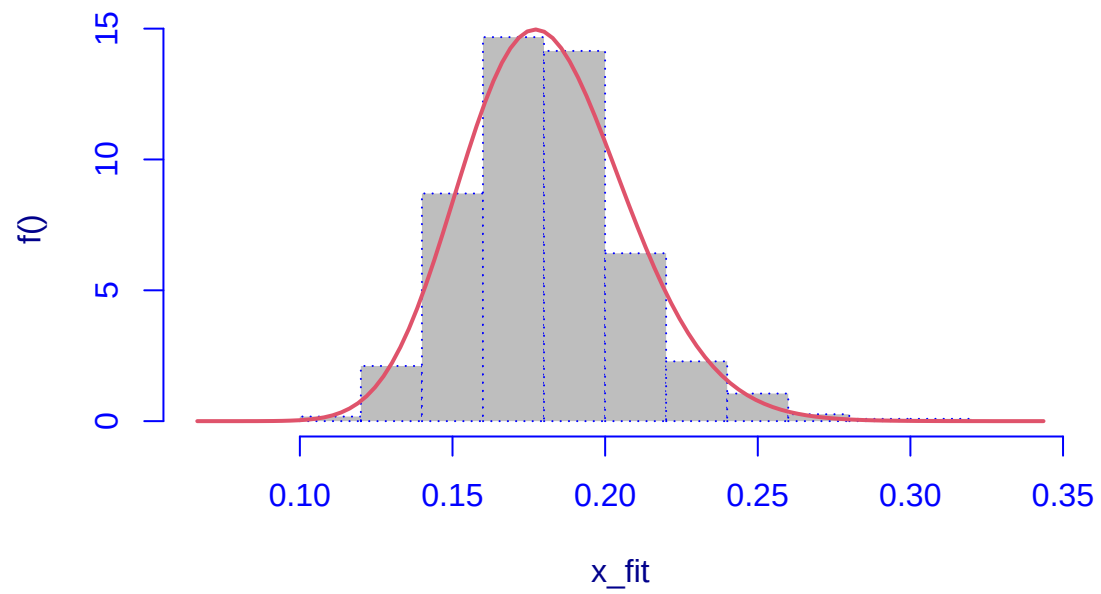
Histogram & Density



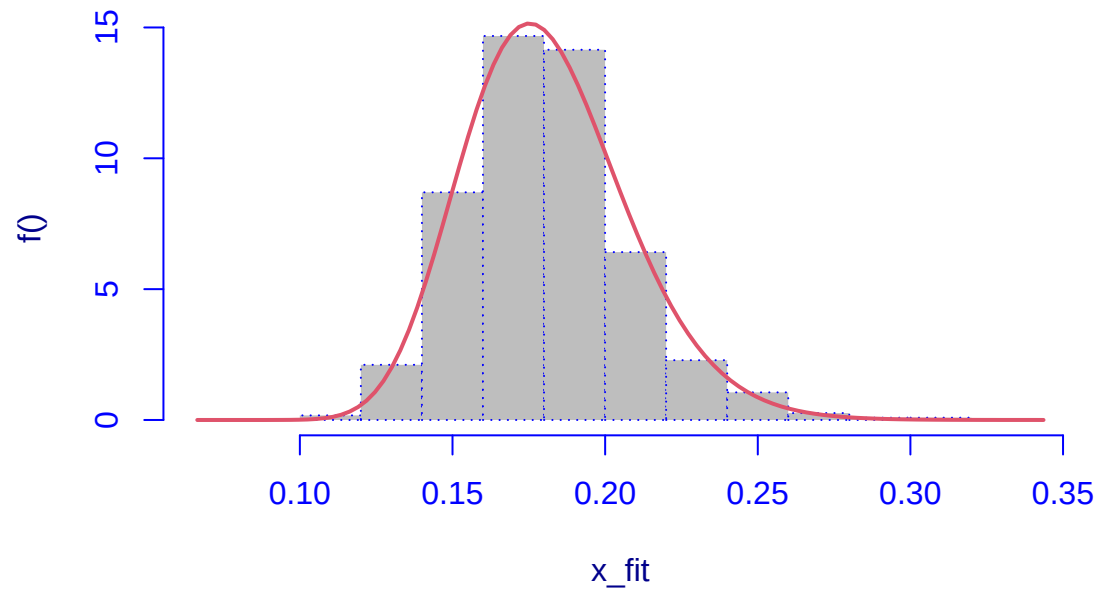
The x_{fit} and the fitted EXP distribution



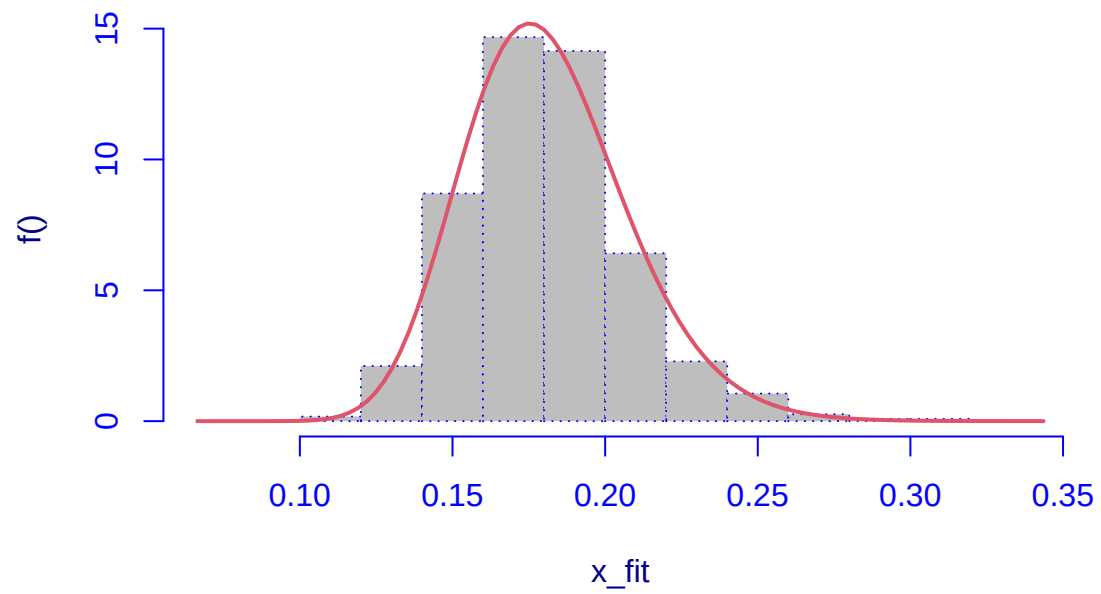
The x_{fit} and the fitted GA distribution



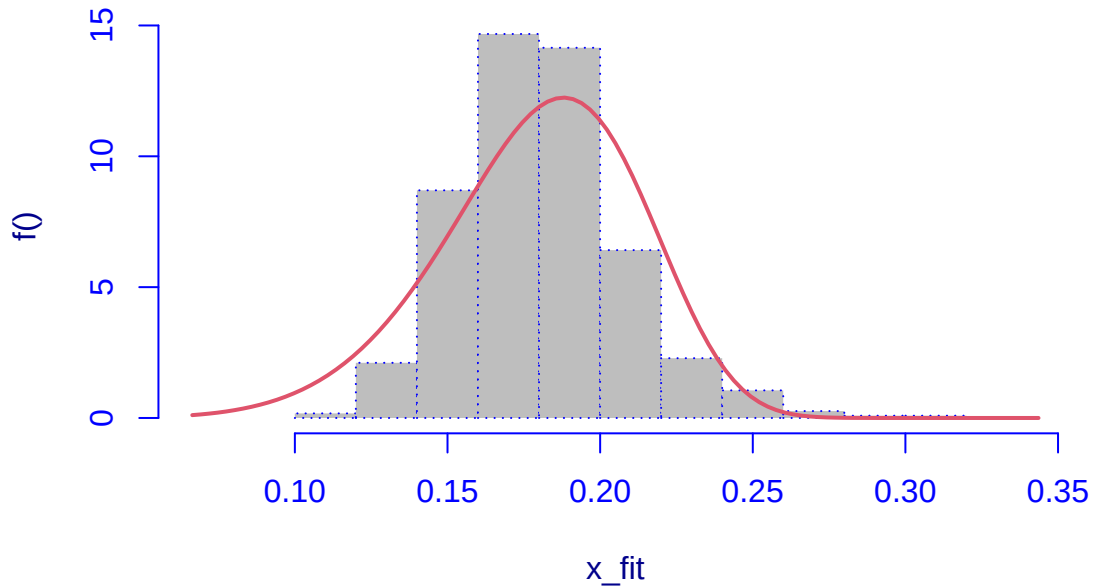
The x_{fit} and the fitted IG distribution



The x_{fit} and the fitted LOGNO distribution



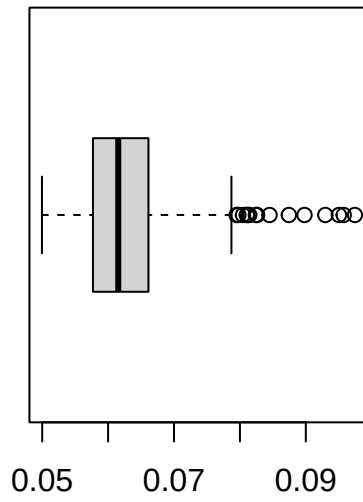
The x_{fit} and the fitted WEI distribution



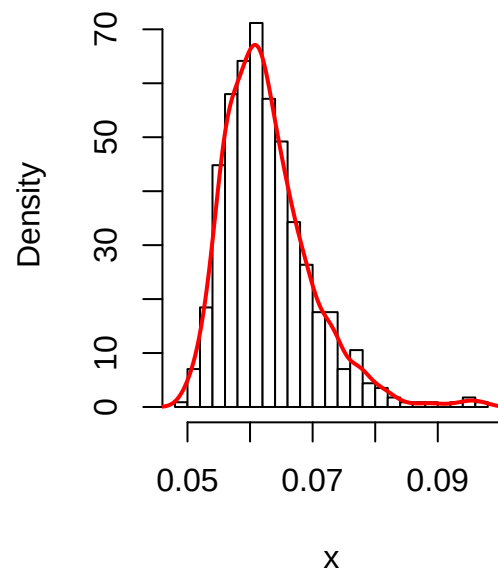
```
##
## **Model Comparison (Sorted by AIC)**
##
##
## |Distribution |      AIC|      BIC|
## |:-----:|:-----:|:-----:|
## |LOGNO      | -2511.40| -2502.71|
## |IG         | -2510.93| -2502.25|
## |GA         | -2504.19| -2495.50|
## |WEI        | -2373.88| -2365.19|
## |EXP        |  -804.12|  -799.77|

##
## **Goodness of Fit (Kolmogorov-Smirnov)**
##
## - **Best Model**: LOGNO
## - **p-value**: 0.6331
## - **Result**: Do NOT reject H0. The model is a good fit.
##
## \newpage
##
## ## Variable: fractal_dimension_mean
##
## **Descriptive Statistics**
##
##
## |  Mean|      SD| Skewness| Kurtosis|
## |:-----:|:-----:|:-----:|:-----:|
## | 0.0628| 0.0071|  1.2976|  2.9481|
```

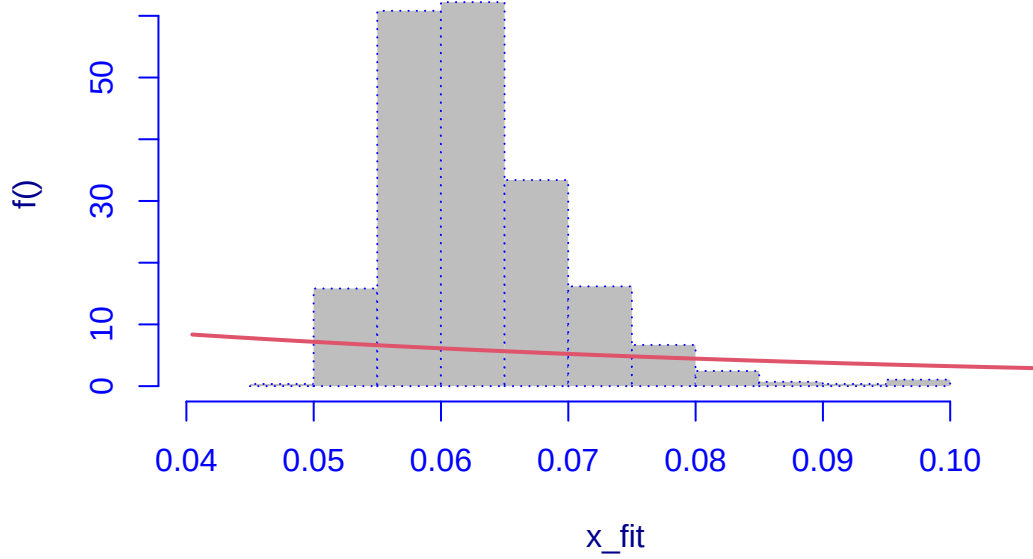
Boxplot



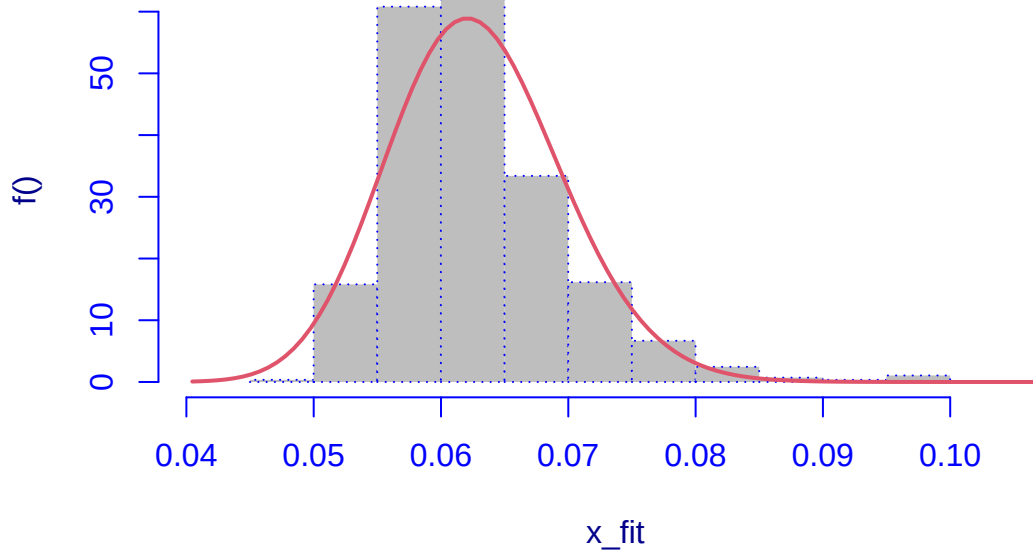
Histogram & Density



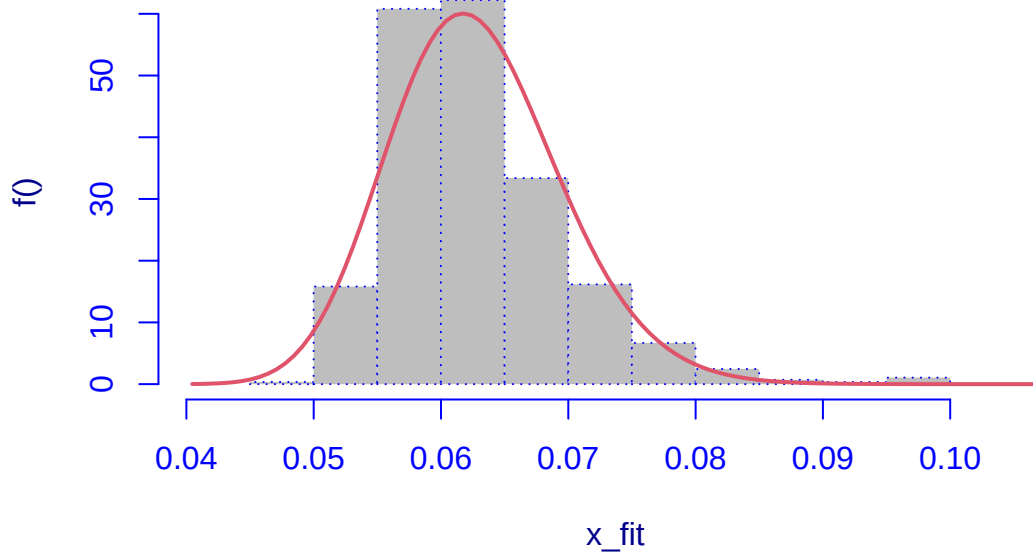
The x_{fit} and the fitted EXP distribution



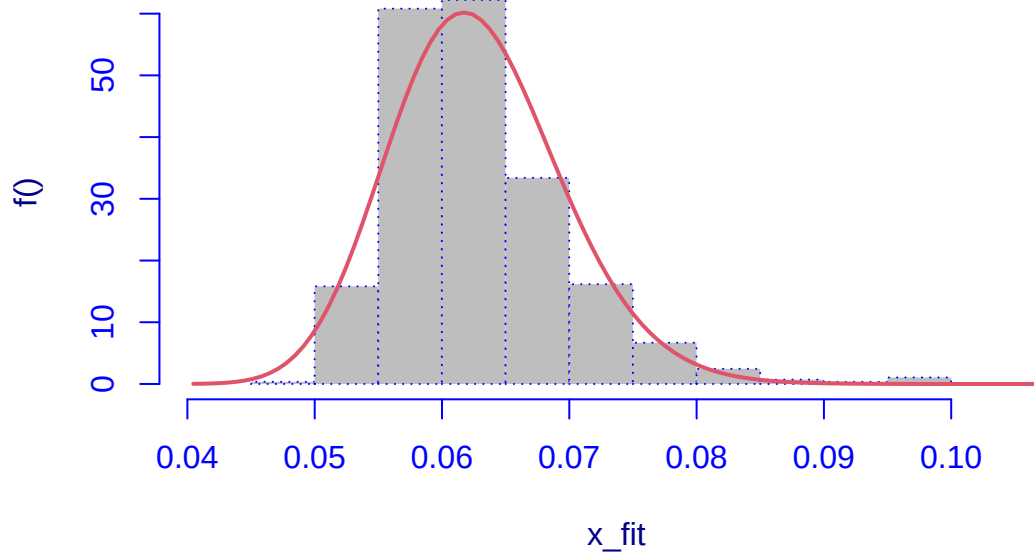
The x_{fit} and the fitted GA distribution



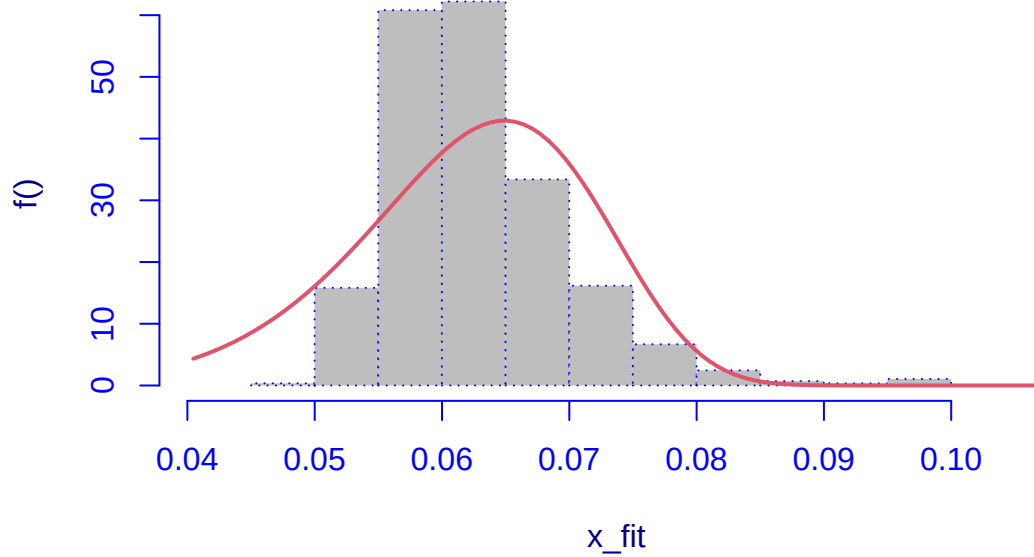
The x_{fit} and the fitted IG distribution



The x_{fit} and the fitted LOGNO distribution



The x_{fit} and the fitted WEI distribution



```
##
## **Model Comparison (Sorted by AIC)**
##
##
## |Distribution |      AIC|      BIC|
## |:-----:|-----:|-----:|
```

```
## |LOGNO          | -4083.50| -4074.81|
## |IG             | -4082.91| -4074.22|
## |GA             | -4064.71| -4056.03|
## |WEI            | -3832.17| -3823.49|
## |EXP            | -2009.80| -2005.46|

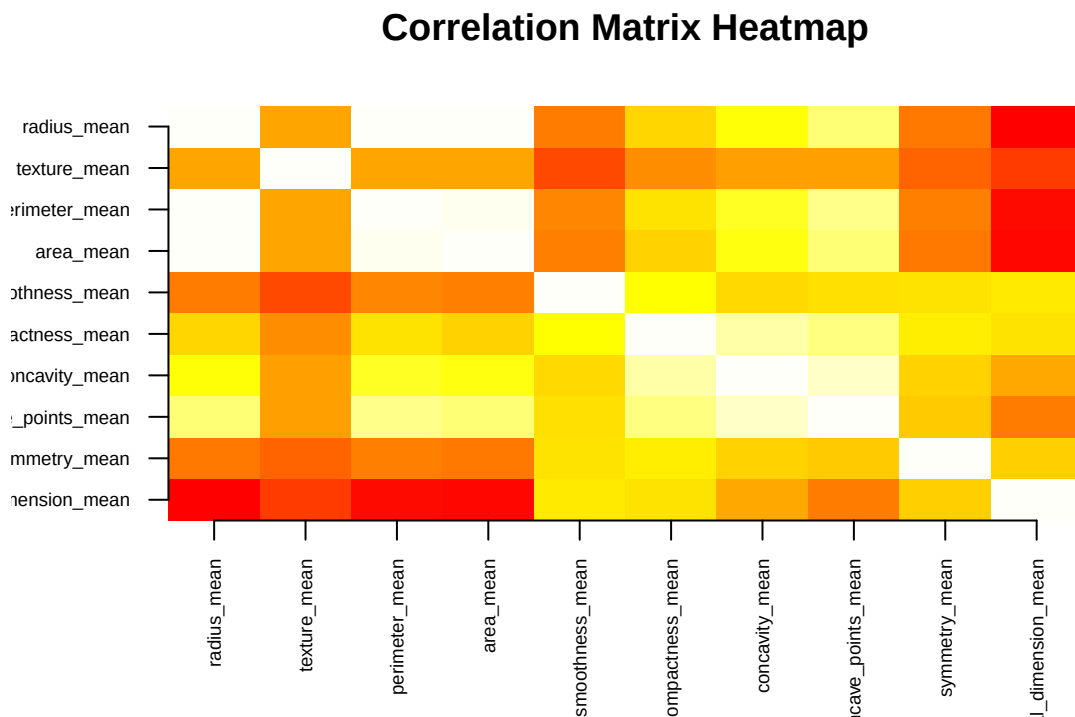
##
## **Goodness of Fit (Kolmogorov-Smirnov)**
##
## - **Best Model**: LOGNO
## - **p-value**: 0.006264
## - **Result**: Reject H0. Significant discrepancy detected.
##
## \newpage
```

2 Principal Component Analysis (PCA)

We apply PCA to reduce dimensionality and handle multicollinearity among the morphological features.

2.1 Correlation Matrix

Before PCA, we inspect the correlation between variables.



High correlations (lighter colors) are observed between radius, perimeter, and area, justifying the use of PCA.

2.2 PCA Results and Component Selection

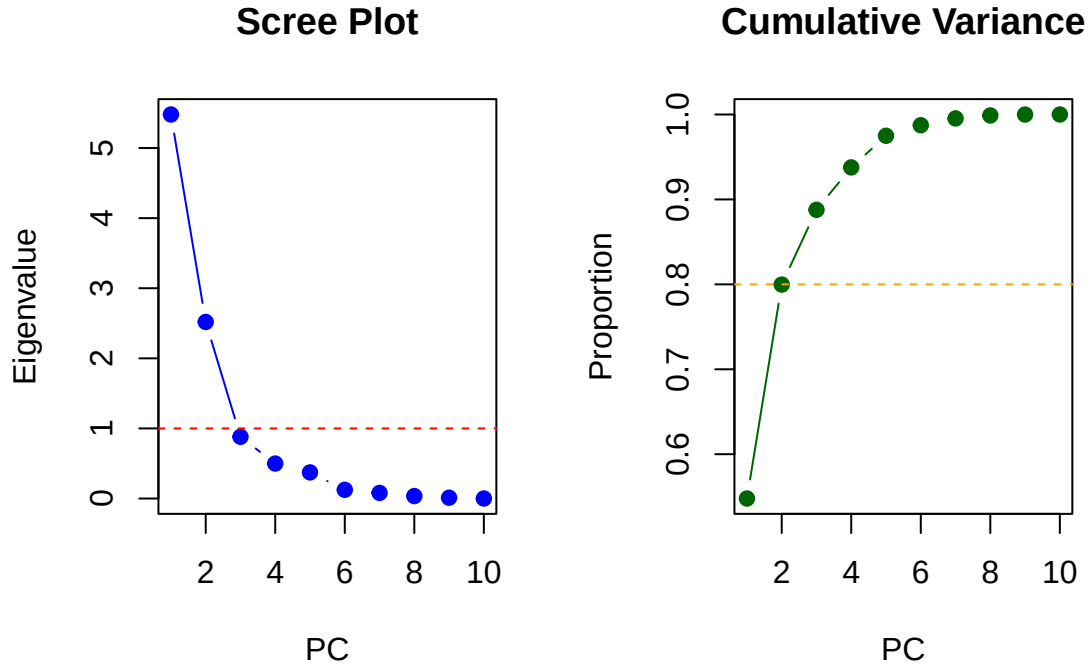


Table 1: Eigenvalues and Explained Variance

Component	Eigenvalue	Variance_Prop	Cumulative_Prop
PC1	5.4786	0.5479	0.5479
PC2	2.5187	0.2519	0.7997
PC3	0.8806	0.0881	0.8878
PC4	0.4990	0.0499	0.9377
PC5	0.3725	0.0373	0.9749
PC6	0.1241	0.0124	0.9874
PC7	0.0801	0.0080	0.9954
PC8	0.0349	0.0035	0.9989
PC9	0.0111	0.0011	1.0000
PC10	0.0003	0.0000	1.0000

Selection Criteria:

* Kaiser's Rule: Keep PCs with Eigenvalue > 1 .

* Variance Explained: Keep PCs describing at least 80% of variance.

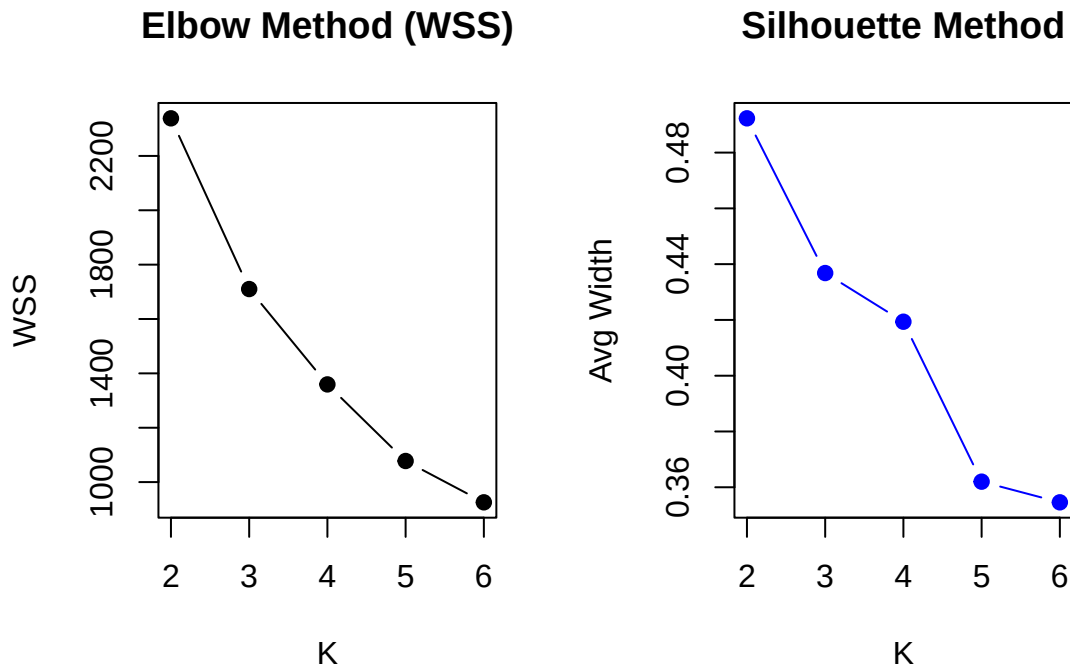
Based on Kaiser's Rule, we select the first 2 components.

3 Cluster Analysis

We perform clustering on the Principal Components (PC1 to PC 2) to identify subgroups in the data.

3.1 Cluster Validation (Choice of K)

We use the Silhouette Method to find the optimal number of clusters (k).

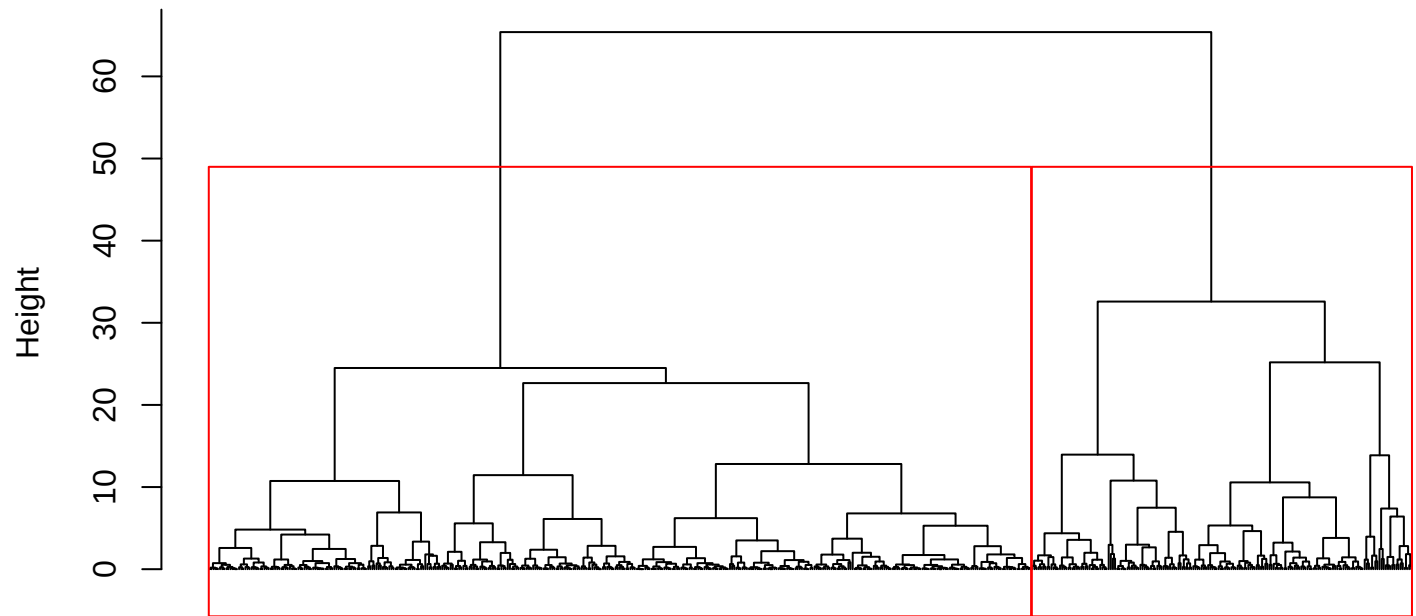


The optimal number of clusters suggested by the Silhouette index is $k = 2$.

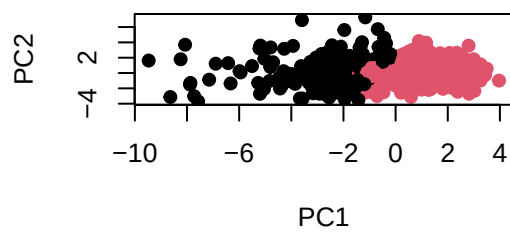
3.2 Clustering Results

We apply Hard (K-Means, Hierarchical) and Soft (Fuzzy) clustering algorithms using $k = 2$. Plots are projected on the first two Principal Components.

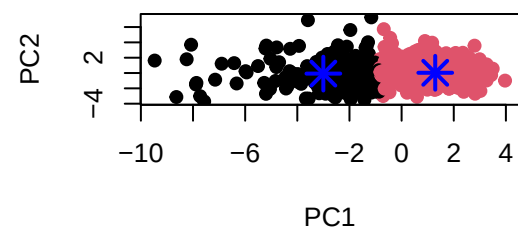
Dendrogram (Ward)



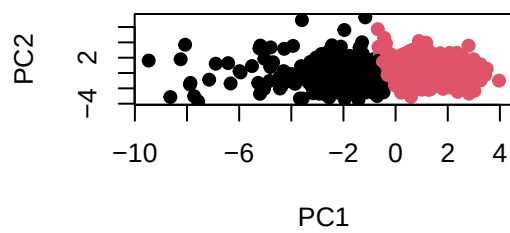
Hierarchical (Ward)



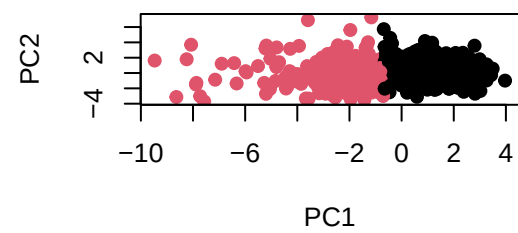
K-Means



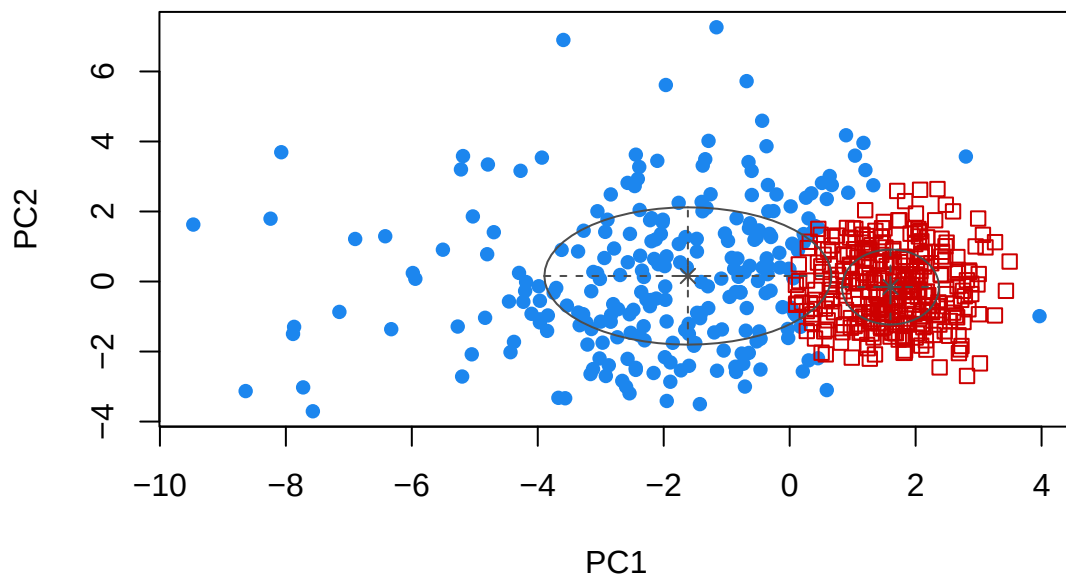
K-Medoids (PAM)



Fuzzy C-Means



3.3 Gaussian Mixture Models (GMM)



```
##
## **GMM Summary:**

## -----
## Gaussian finite mixture model fitted by EM algorithm
## -----
##
## Mclust VVI (diagonal, varying volume and shape) model with 2 components:
##
## log-likelihood  n df      BIC      ICL
##      -2226.677 569  9 -4510.449 -4654.213
##
## Clustering table:
##   1  2
## 253 316
```